



Power BI

FUNDAMENTALS

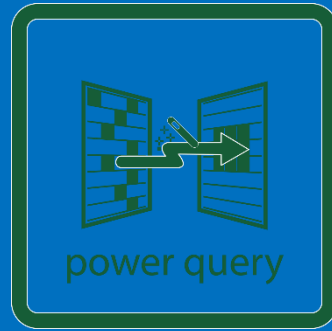
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POWER
BI



POWER
QUERY



POWER
PIVOT



DAX

SUMMARY



WHAT IS POWER BI ?



Power BI
Desktop



POWER BI DESKTOP



Power BI
Desktop



 **POWER QUERY**



Power BI
Desktop



DATA MODELING
Relational database Concepts



Power BI
Desktop



DAX
Data Analysis Expressions



Power BI
Desktop



POWER VIEW
Visuals in Power BI



Power BI
Desktop



 **POWER BI SERVICE**
Dashboards and sharing



Power BI
Desktop



POWER BI: BEST PRACTICES



Power BI
Desktop



POWER BI: PRICING & other products



Power BI
Desktop



WHAT IS POWER BI ?



Power BI

Desktop

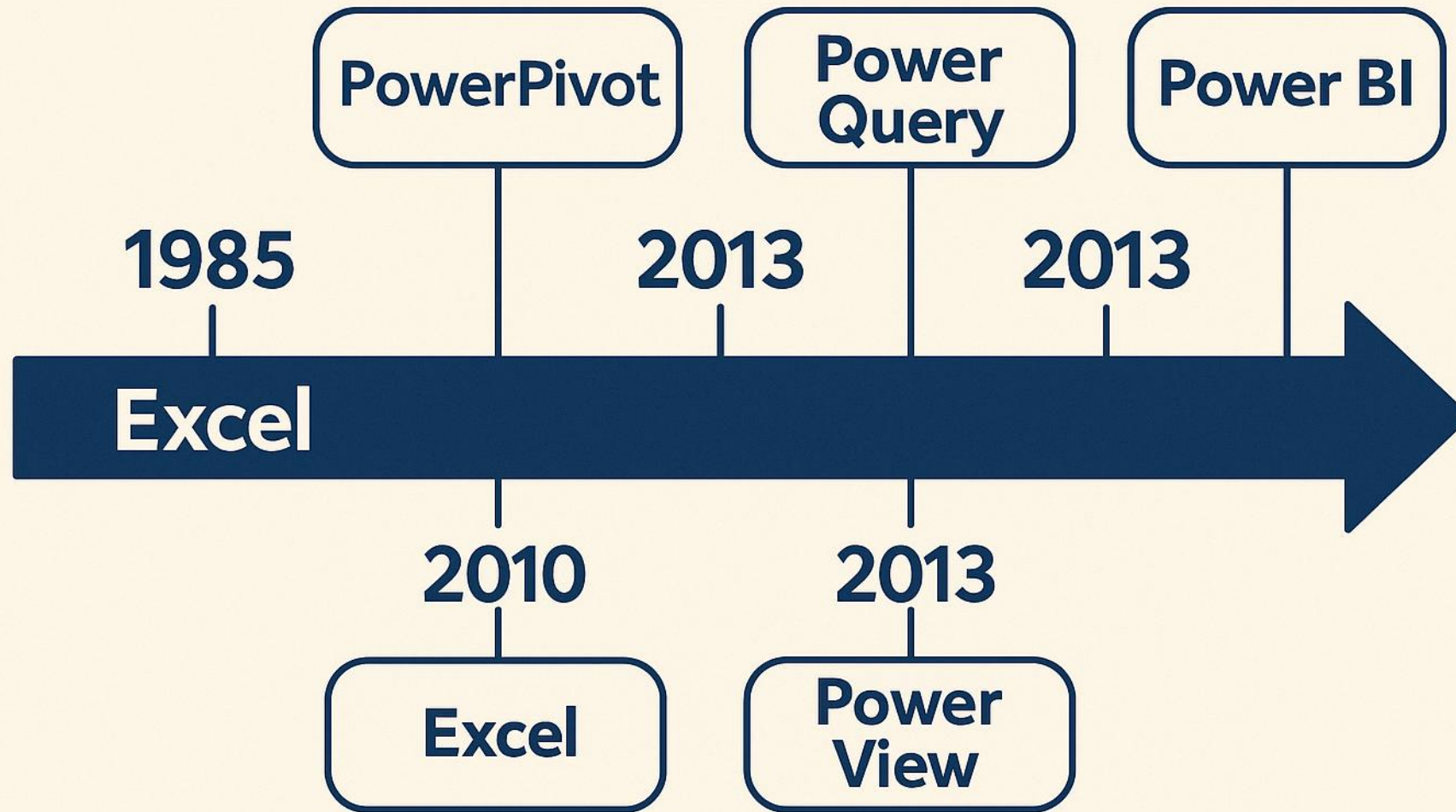


Power BI – BUSINESS INTELLIGENCE Made Easy

Power BI is Microsoft's data visualization and business intelligence tool that enables users to:

- Connect to various data sources (Excel, databases, cloud services)
- Transform and model data with ease
- Create interactive dashboards and reports
- Share insights across teams and devices

A little bit of history



Power BI vs Excel

	Power BI	Excel
Data connectors available	200+	50+
Different visualizations available	400+	40+
Filtering at the report level or the page level	YES	NO
Mobile layout optimization available in power BI	YES	NO
Drill through on charts and cross-filtering	YES	NO
Formula language	DAX	Excel
R and Python languages integration	NATIVE	LIMITED
Versioning	PBI service	NO
Powerpoint interactive report or chart integration	YES	LIMITED
Power View	NATIVE	DISCONTINUED
ISO 8000 compatible	YES	NO

The 3 primary components of Power BI

Power BI Desktop



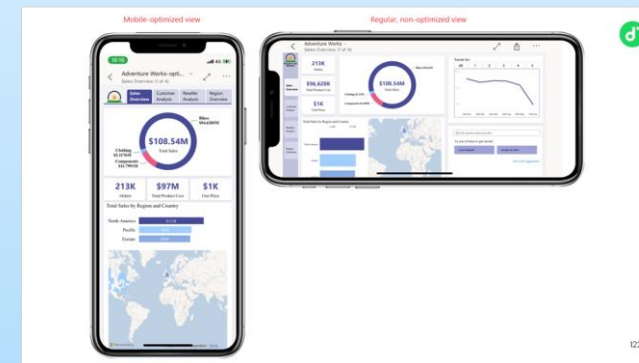
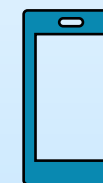
is the free development tool available to data analysts and other report creators

Power BI Service



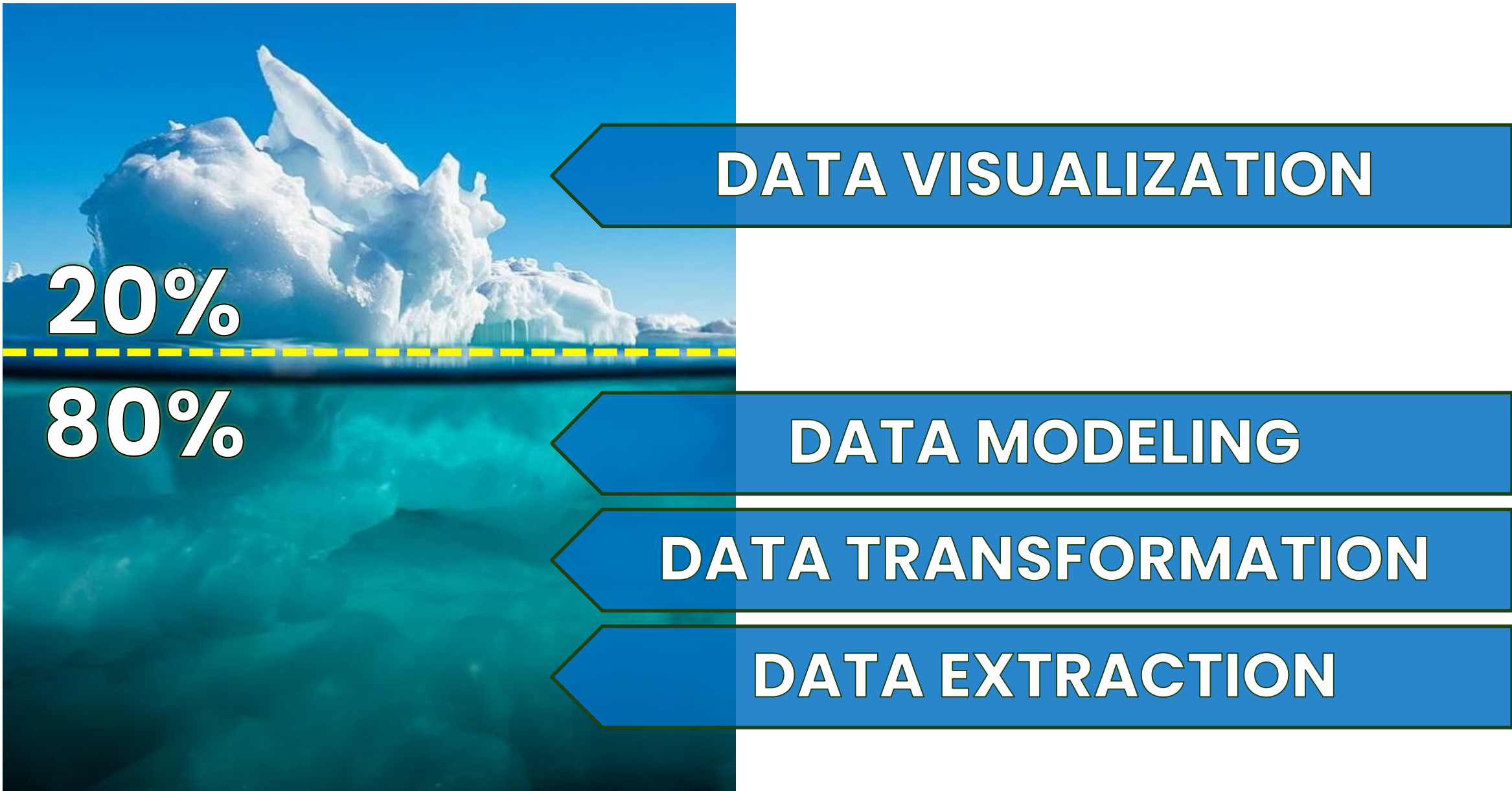
allows you to organize, manage, and distribute your reports and other Power BI items.

Power BI Mobile



allows consumers to view reports in a mobile-optimized format.

DATA ICEBERG MODEL



Key Features of Business Intelligence Tools

REPORTING

#flexible
#customizable
#perspectives

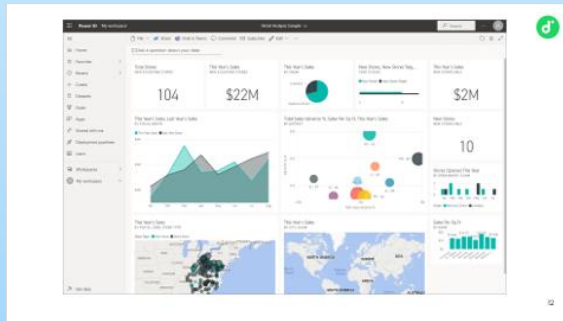


A screenshot of a BI reporting tool displaying a table of sales data. The table has columns for Category, This Year Sales Status, Average Unit Price, Last Year Sales, This Year Sales, and This Year Sales Goal. The data is categorized by product types like 010-Womens, 020-Mens, etc., and includes a total row at the bottom.

Category	This Year Sales Status	Average Unit Price	Last Year Sales	This Year Sales	This Year Sales Goal
010-Womens	●	\$7.30	\$2,680,662	\$1,787,958	\$2,680,662
020-Mens	●	\$7.12	\$4,453,133	\$4,452,421	\$4,453,133
030-Kids	●	\$5.30	\$2,726,892	\$2,705,490	\$2,726,892
040-Juniors	●	\$7.00	\$3,105,550	\$2,930,385	\$3,105,550
050-Shoes	●	\$13.84	\$3,640,471	\$3,574,900	\$3,640,471
060-Intimate	●	\$4.28	\$955,370	\$852,329	\$955,370
070-Hosiery	●	\$3.69	\$573,604	\$486,106	\$573,604
080-Accessories	●	\$4.84	\$1,273,096	\$1,379,259	\$1,273,096
090-Home	●	\$3.93	\$2,913,647	\$3,053,326	\$2,913,647
100-Groceries	●	\$1.47	\$810,176	\$829,776	\$810,176
Total	●	\$5.49	\$23,132,601	\$22,051,952	\$23,132,601

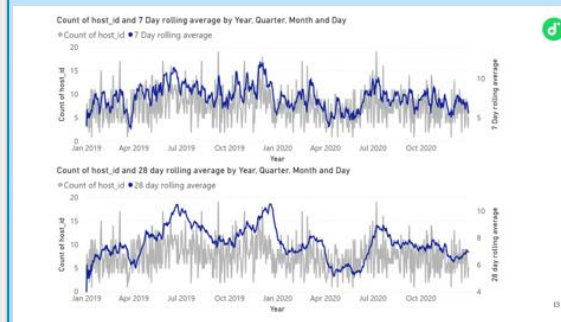
DATA VISUALIZATION

#intuitive
#interactive
#understandable



DATA MINING

#patterns
#relationships
#trends

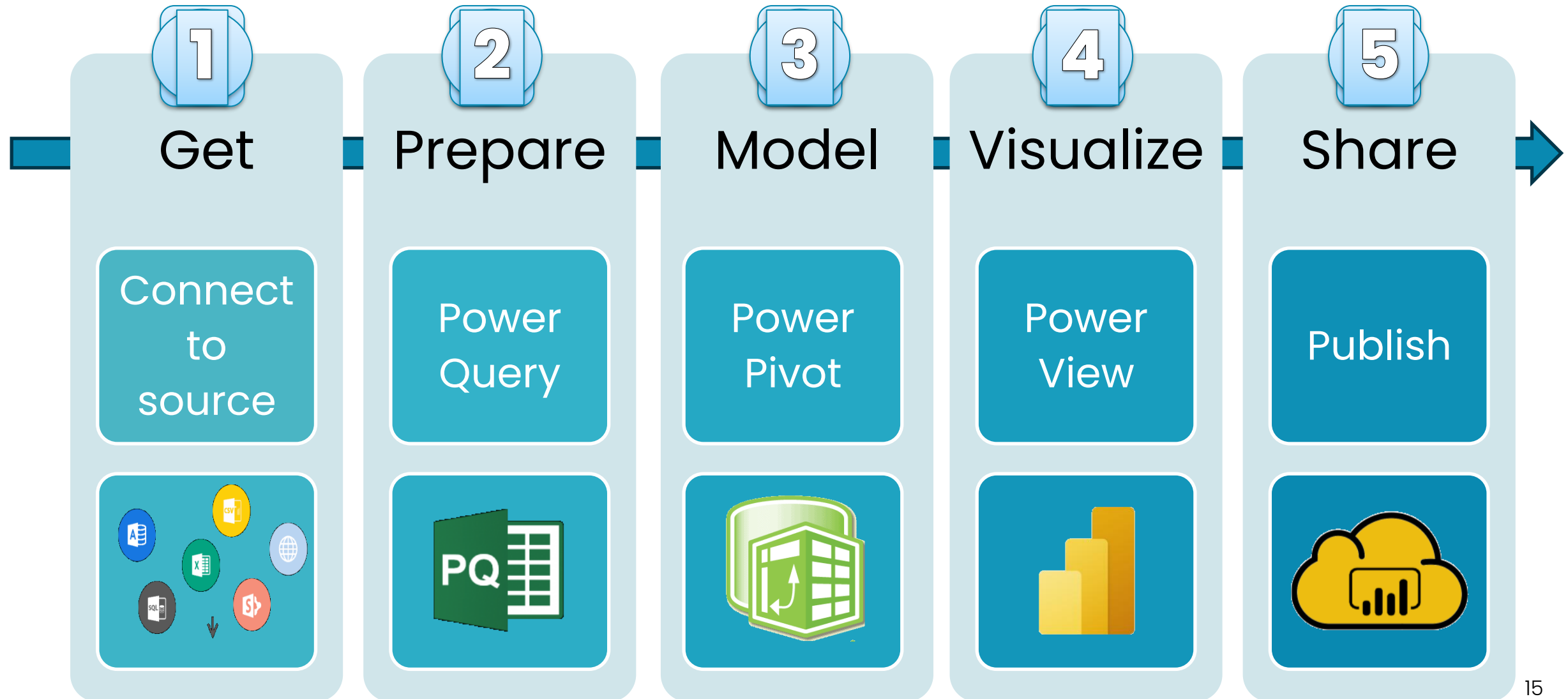


DASHBOARDS

#realtime
#kpi
#monitor
#performance



Power BI data flow



Recommendations

- Edit your PBI reports in the desktop app (not PBI service).
- Save regularly, even when using cloud stored files with auto save
- Do not neglect data modeling or data normalization
- Do not think like Excel
- Optimize queries and don't hesitate to transform data
- Don't underestimate DAX, use it ...
- Choose a license that is suited for your company use



POWER BI DESKTOP



Power BI

Desktop



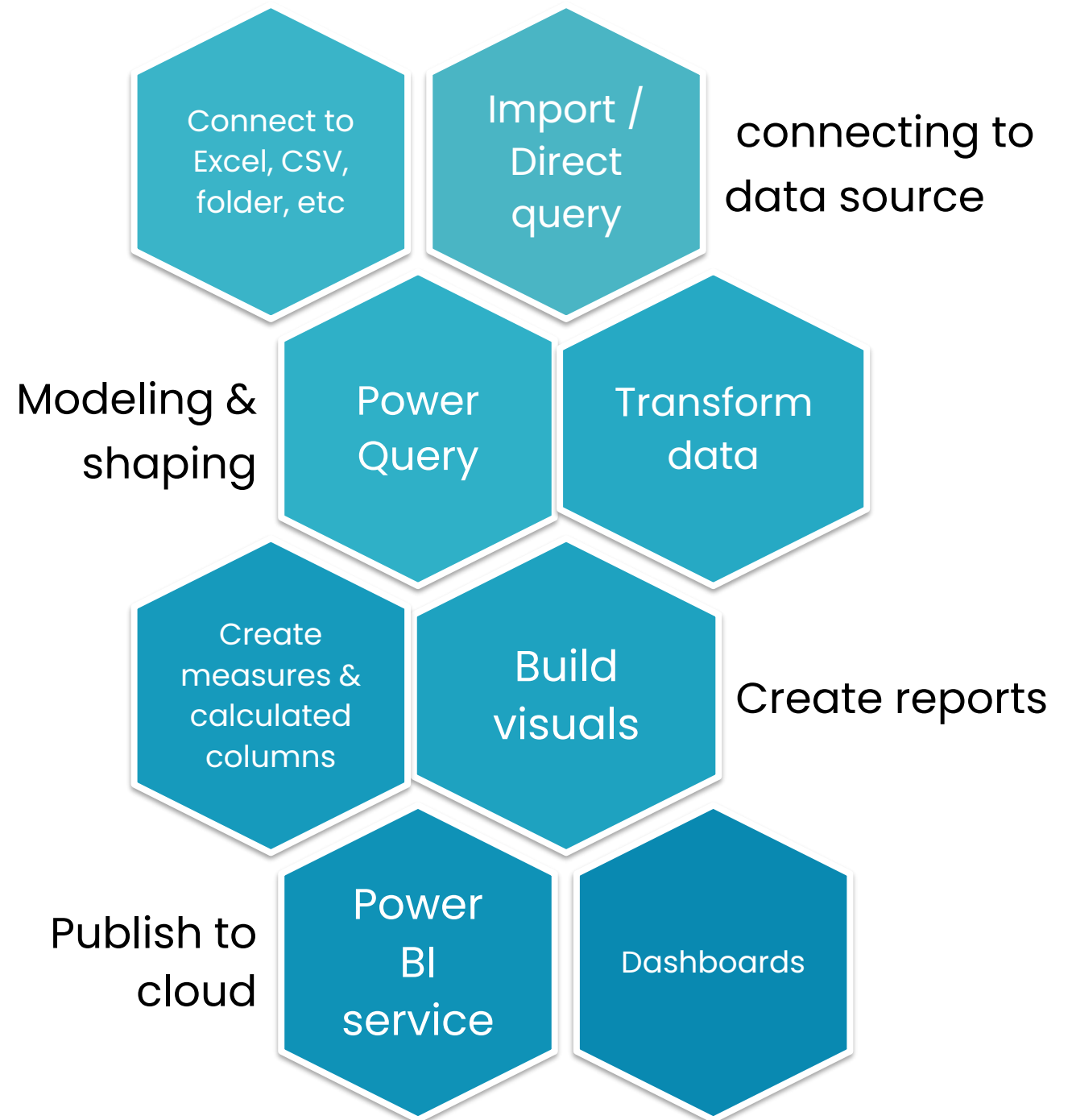


Power BI

DESKTOP



- Create semantic models and visuals to build reports.



Semantic model = connected data, transformations, relationships and calculations.

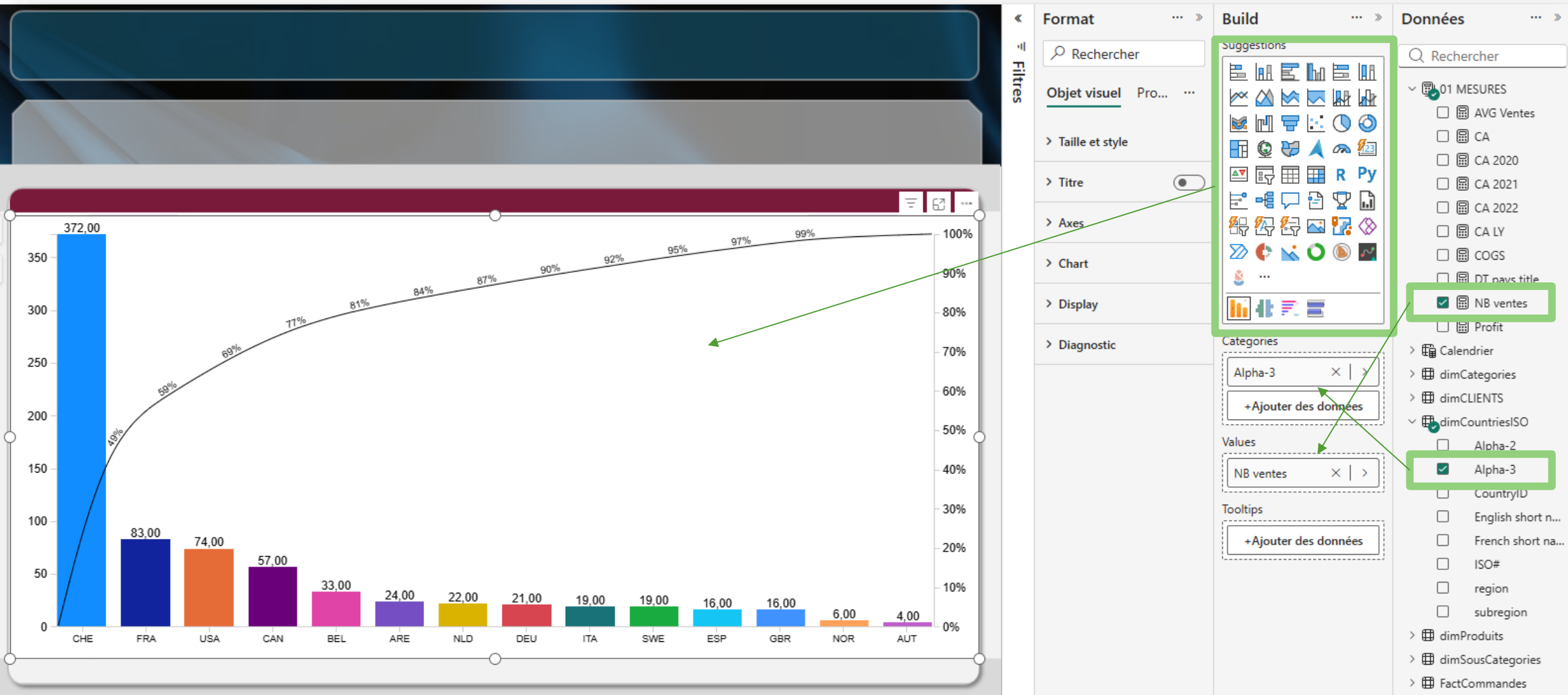
Introducing power BI desktop

The image shows the Power BI Desktop interface with several components labeled by blue arrows and boxes:

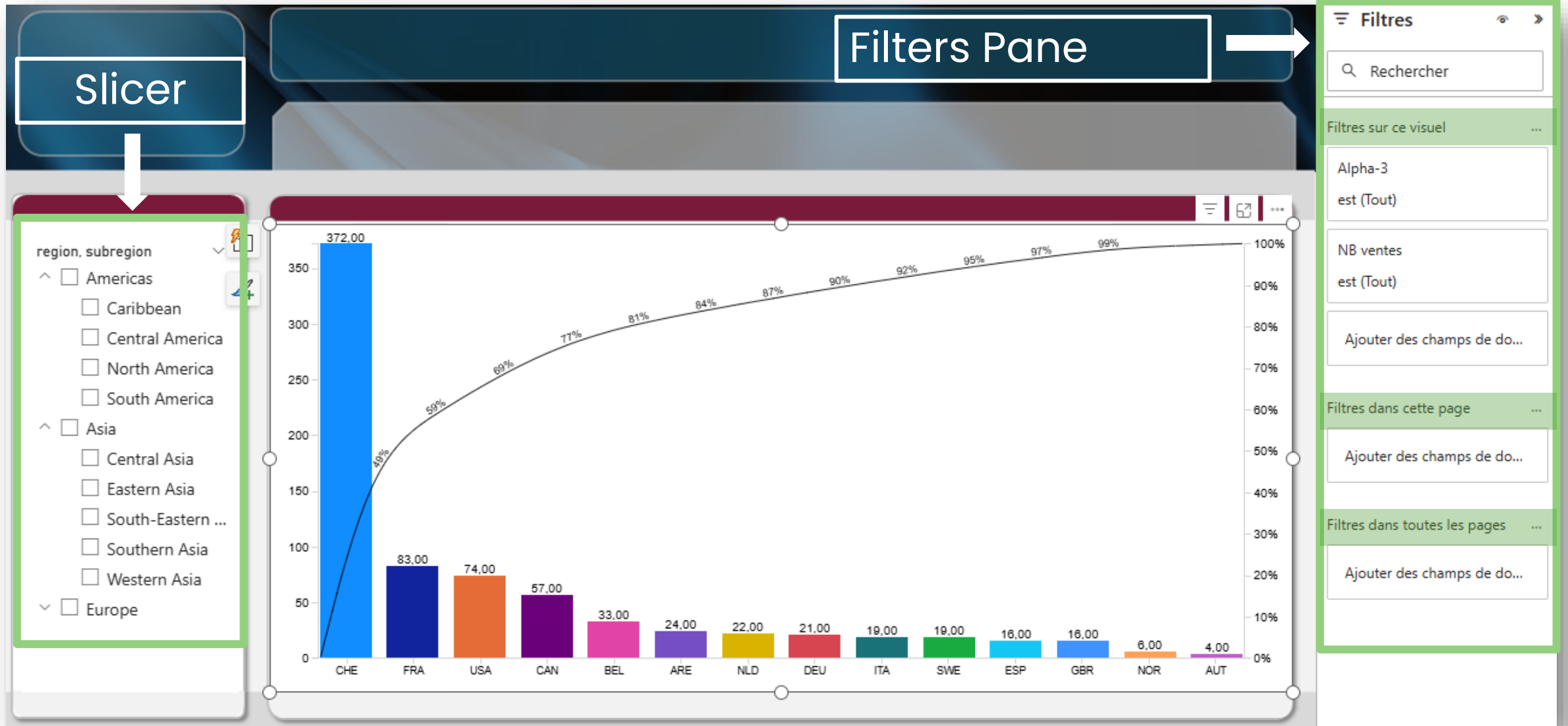
- Quick access**: Points to the top-left corner of the ribbon.
- Ribbons**: Points to the ribbon tabs (File, Home, Insert, Modeling, View, Optimize, Help).
- Report view Data view Model view**: Points to the left-hand navigation pane.
- Canvas**: Points to the central workspace area.
- Page tabs**: Points to the bottom-left page navigation area.
- Status bar**: Points to the bottom status bar.
- Filters pane**: Points to the top-left pane of the right-hand sidebar.
- Visualizations pane**: Points to the top-middle pane of the right-hand sidebar.
- Data pane**: Points to the top-right pane of the right-hand sidebar.

The interface includes a ribbon with tabs: File, Home, Insert, Modeling, View, Optimize, and Help. The Home tab is active, showing options like Paste, Cut, Copy, Format painter, Clipboard, Get data, Excel workbook, OneLake data hub, SQL Server, Enter data, Dataverse, Recent sources, Transform data, Refresh data, New visual, Text box, More visuals, New measure, Quick measure, Sensitivity, and Publish. The left-hand navigation pane shows Report view, Data view, and Model view. The central canvas displays "Add data to your report" with options: Import data from Excel, Import data from SQL Server, Paste data into a blank table, and Try a sample dataset. The right-hand sidebar has three panes: Filters (with search and filters on this/all pages), Visualizations (with search, build visual, and various chart types), and Data (with search and a message "You haven't loaded any data yet. Get data"). The bottom status bar shows "Page 1 of 1" and a zoom level of 60%.

Create a visual



Filter a visual



Creating a report (.pbix file)

Connect to data
& populate the
data model.

Add DAX tables,
columns or
measures.

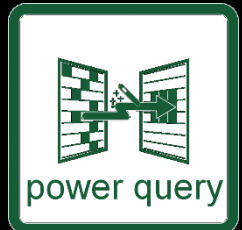
Create and
format visuals.

Insert Slicers for
dynamic
filtering.

Create
Drillthrough and
tooltip pages.

Apply
conditional
formatting.

Use bookmarks
to create
snapshots.

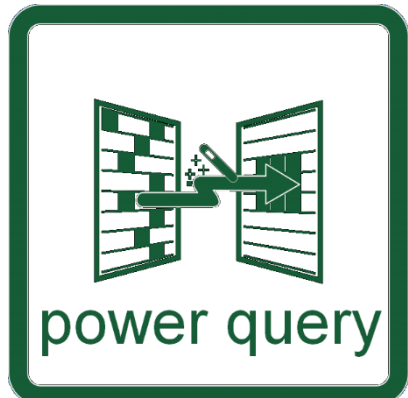


POWER QUERY



Power BI
Desktop





Seamlessly connect to multiple sources

Easily reshape, cleanse, combine and merge data.

Unpivot data.

Create calendar tables.

Add calculated columns.

Agregates data

Automate & reuse queries in other files.

Data Sources: File-based connectors



Excel (.xlsx, .xls)

Most frequent source for self-service BI

CSV / Text

Flat file data dumps, exports from ERP/CRM systems

XML / JSON

APIs, configurations, structured exports

Folder

Batch import of multiple files with the same structure

Data Sources: Database connectors



SQL Server	Azure SQL DB	Oracle	MySQL / PostgreSQL	Access	SAP HANA / BW
Core enterprise database, supports Import & DirectQuery	Cloud-based SQL platform	Legacy ERP & financial data	Common in open-source or web-based apps	Still used in legacy environments	Business-critical SAP system data

Data Sources: Database connectors



SharePoint / OneDrive

Excel/CSV in collaborative lists/folders. Auto-syncing cloud files.

Google Analytics / BigQuery

Web & marketing data platforms

Microsoft 365

Email, Teams, Planner analytics

Dynamics 365

CRM/ERP sales, ops data

Azure Data Lake / Blob

Cloud data storage and staging

Salesforce

CRM leads, accounts, opportunities

Data Sources: Database connectors



Web

Extract data from websites or APIs

ODBC / OLE DB

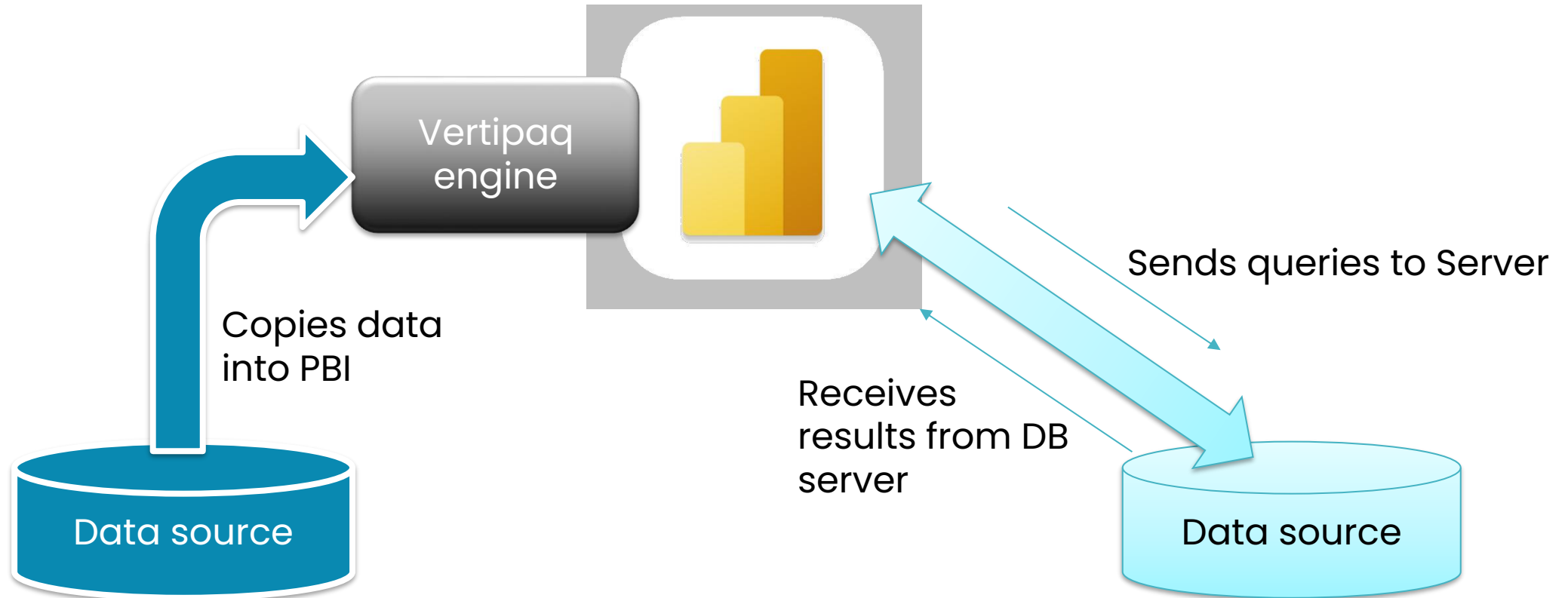
Generic access to niche/proprietary systems

R / Python

Embed scripts for advanced ETL

Blank Query

Manual entry of M logic and custom transformations



Not available for all sources or for certain visuals.
Sensitive to remote server efficiency.

DirectQuery

Data sources that support DirectQuery.

(Sensitive to remote server efficiency.)

- SQL Server (on-prem)
- Azure SQL Database
- Azure SQL Managed Instance
- Azure Synapse Analytics
- Oracle
- Teradata
- SAP HANA
- Snowflake
- Google BigQuery
- Amazon Redshift
- PostgreSQL

You can also create a composite model to get the best of both worlds:

Large Fact Table -> DirectQuery

Small Dimension Tables -> Import

Avoid DirectQuery when:

- Users expect very fast interactions
- Reports contain complex DAX
- Many slicers, cross-filtering, drill-downs
- Source database is not optimized for analytics
- You need:
 - Complex calculated columns
 - Many time intelligence measures
 - High concurrency

Launch Power Query Editor

The screenshot shows the Power Query Editor interface. On the left, the 'Get Data' pane is open, showing various data sources. A red box highlights the 'Get data' button in the 'Home' ribbon. A large grey arrow points from this button to the main editor window. In the main editor window, a red box highlights the 'Data source settings' button in the 'Home' ribbon. Another large grey arrow points from this button to the 'Data source settings' button. The main editor window displays a table of data with columns 'Segment', 'Country', and 'Product'. A red box highlights the 'Data source settings' button in the 'Home' ribbon. The 'Query Settings' pane on the right shows the 'Properties' tab with the query name 'financials' and the 'Applied Steps' list containing 'Source', 'Navigation', and 'Changed Type'. Red circles with numbers 1 through 4 are placed over the 'Data source settings' button, the 'financials' query name, the 'Data source settings' button, and the 'Properties' tab, respectively.

1. Ribbons (tools)

2. Queries

3. Data preview window

4. Properties & applied steps



Transforming data using Power query

Columns

Insert / delete

Combine / split

Calculate

Unpivot

Filter / Sort

Rename

Rows

Remove duplicates

Remove blanks

Filter out

Group

Transpose

Text

Replace

Format

Fill blanks

Trim

Extract

Cleanup with Power Query

- Remove useless rows and columns
- Promote headers in first row
- Split columns (i.e. the Dimension col here).
- Check and correct data types if necessary.
- Import and convert different character encoded files (ANSI, UTF, etc)
- Load the Data into the Data model

	ABC Column1	ABC Column2	ABC Column3
1	Voici un tableau avec 50 noms de produits inform...	null	null
2	null	null	null
3	Noms de produit Poids (kg) Dimensions ...	null	null
4	----- ----- -----	null	null
5	Ordinateur portable 2.5 35 x 24 x 2...	null	null
6	Moniteur 5.2 60 x 40 x 1...	null	null
7	Clavier 0.8 45 x 15 x 3...	null	null
8	Souris 0.1 10 x 6 x 3 ...	null	null
9	Imprimante 7.8 45 x 35 x 2...	null	null
10	Scanner 3.2 30 x 20 x 1...	null	null
11	Routeur 0.4 20 x 15 x 5...	null	null
12	Modem 0.3 15 x 10 x 4...	null	null
13	Disque dur externe 0.2 10 x 8 x 2 ...	null	null
14	Cl? USB 0.02 5 x 2 x 0.5...	null	null
15	Carte graphique 1.2 25 x 10 x 4...	null	null
16	Carte m?re 0.6 30 x 20 x 3...	null	null
17	Processeur (CPU) 0.1 5 x 5 x 2 ...	null	null
18	M?moire RAM 0.05 8 x 2 x 0.5...	null	null
19	Disque dur interne 0.4 10 x 7 x 2 ...	null	null
20	SSD (Solid-State Drive) 0.05 5 x 3 x 0.5 ...	null	null
21	cran tactile 3.8 55 x 30 x 5 ...	null	null
22	Casque audio 0.3 20 x 18 x 1...	null	null
23	Haut-parleurs 1.2 25 x 15 x 1...	null	null
24	Webcam 0.1 5 x 5 x 2 ...	null	null
25	Microphone 0.2 10 x 5 x 3 ...	null	null
26	Tablette 0.5 25 x 18 x 1...	null	null
27	Smartphone 0.2 15 x 7 x 0....	null	null
28	Smartwatch 0.1 4 x 4 x 1 ...	null	null
29	couteurs sans fil 0.05 3 x 2 x 1 ...	null	null
30	Routeur Wi-Fi 0.3 15 x 10 x 4...	null	null
31	Adaptateur Bluetooth 0.02 3 x 2 x 1 ...	null	null
32	null	null	null
33		null	null
34	Batterie externe 0.2 10 x 5 x 2...	null	null

Ease of use vs M language

- Beginner: Use visual tools, quick undo with applied steps
Advanced: Use M functions

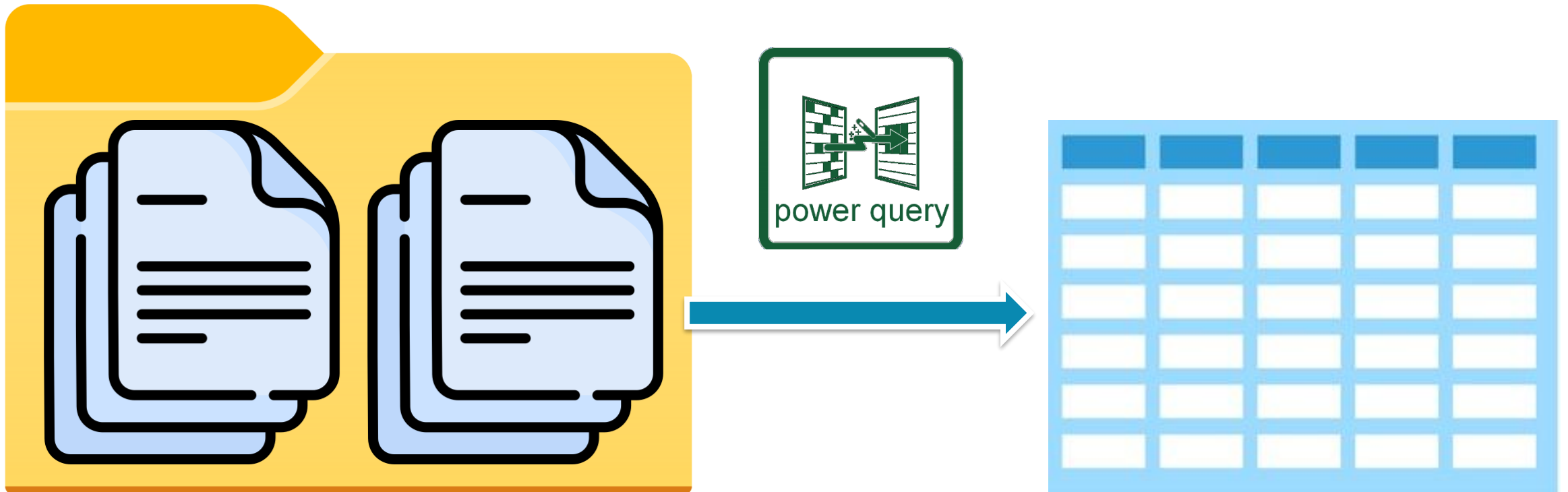
```

let
    Source = Excel.CurrentWorkbook()[[Name="Tableau5"]][Content],
    #"Type modifié" = Table.TransformColumnTypes(Source,{{"Colonne1", type any}, {"Colonne2", type text}, {"Colonne3", type any}, {"Colonne4", type any}}, {"Colonne4", type any}),
    #"Premières lignes supprimées" = Table.Skip(#"Type modifié",3),
    #"Rempli vers le bas" = Table.FillDown(#"Premières lignes supprimées",{"Colonne1"}),
    #"Colonnes fusionnées" = Table.CombineColumns(#"Rempli vers le bas",{"Colonne1", "Colonne2"},Combiner.CombineTextByDelimiter(":", QuoteStyle.None),{"Colonne1"}),
    #"Table transposée" = Table.Transpose(#"Colonnes fusionnées"),
    #"Rempli vers le bas1" = Table.FillDown(#"Table transposée",{"Column1"}),
    #"En-têtes promus" = Table.PromoteHeaders(#"Rempli vers le bas1", [PromoteAllScalars=true]),
    #"Supprimer le tableau croisé dynamique des autres colonnes" = Table.UnpivotOtherColumns(#"En-têtes promus", {"Colonnes fusionnées", "Table transposée"}, "Attribut", "Valeur"),
    #"Fractionner la colonne par délimiteur" = Table.SplitColumn(#"Supprimer le tableau croisé dynamique des autres colonnes", "Attribut", SplitByDelimiter, {"Attribut.1", "Attribut.2"}),
    #"Type modifié2" = Table.TransformColumnTypes(#"Fractionner la colonne par délimiteur",{{"Attribut.1", type text}, {"Attribut.2", type text}}, {"Attribut.2", type text}),
    #"Colonnes renommées" = Table.RenameColumns(#"Type modifié2",{{":", "Statut"}, {"Succursalle:Description", "Mois"}, {"Attribut.1", "Ville"}),
    #"Type modifié3" = Table.TransformColumnTypes(#"Colonnes renommées",{{"Mois", type date}})
in
    #"Type modifié3"

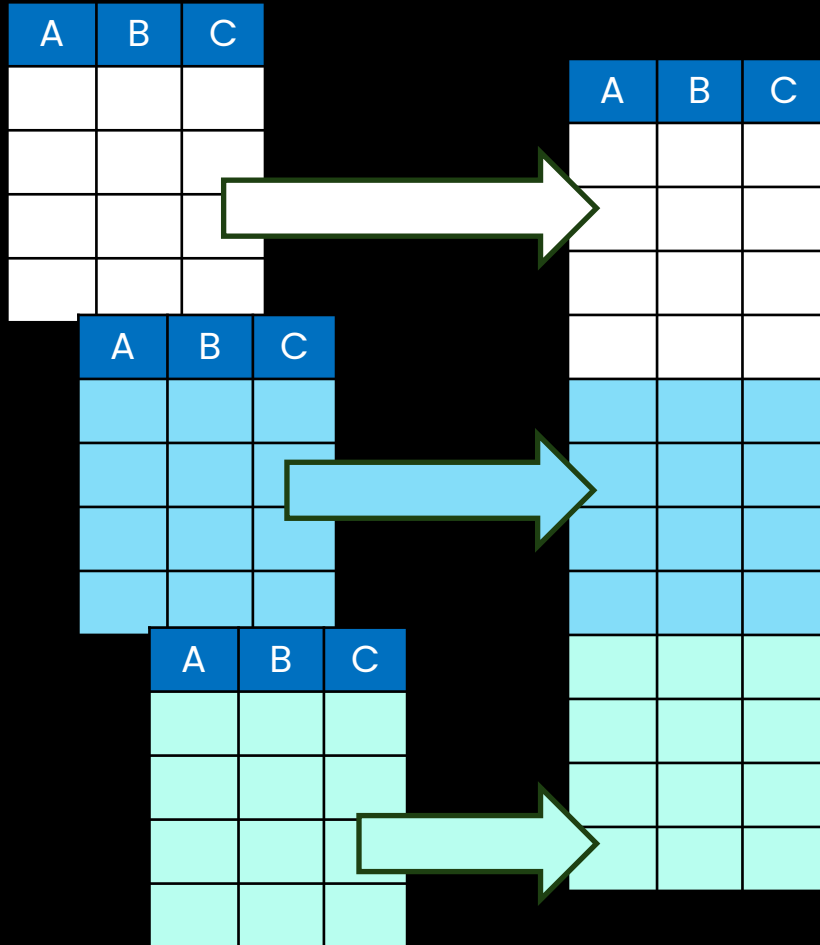
```

Consolidating from a folder

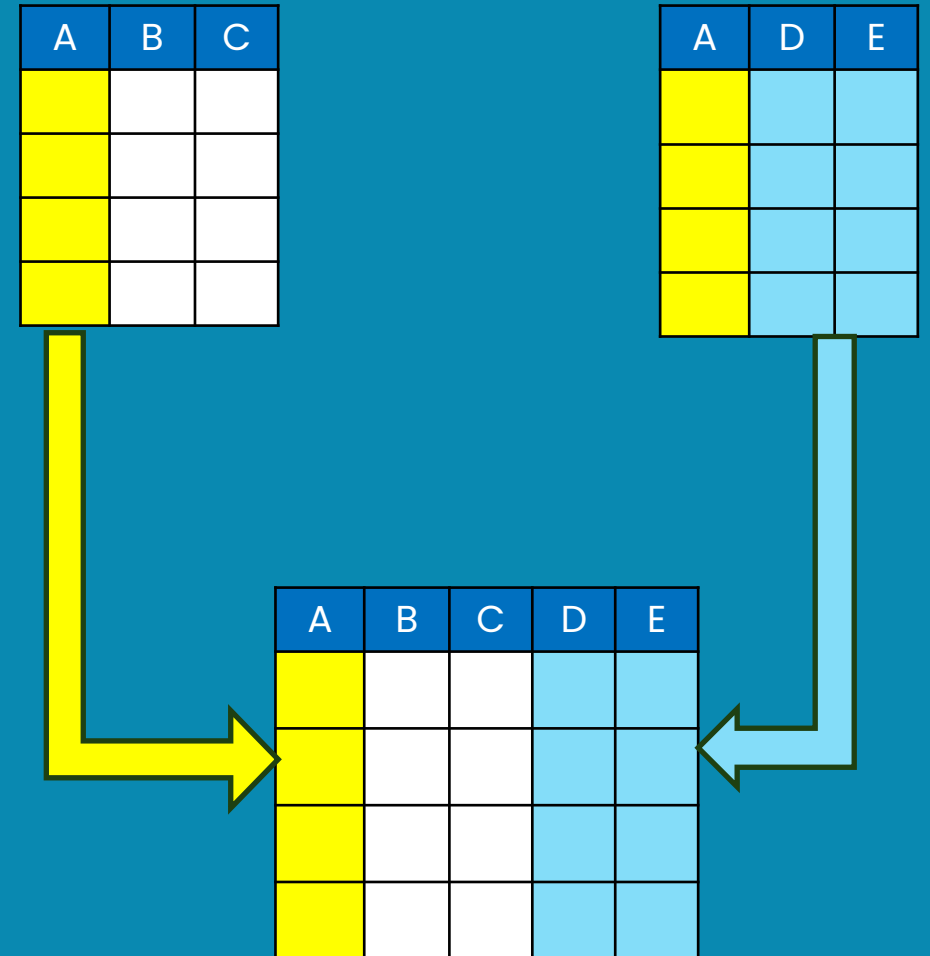
Transform data coming from files inside a folder and easily update with one click whenever new files are added to that folder.



APPENDING



MERGING



MERGE 2 QUERIES



SKU	souscatID
● Valide 100%	● Valide 100%
● Erreur 0%	● Erreur 0%
● Vide 0%	● Vide 0%
SKU-619955	5
SKU-367525	7
SKU-874528	16
SKU-167410	18
SKU-668353	25
SKU-772806	17
SKU-640119	29
SKU-154609	25
SKU-791801	35
SKU-129497	23
SKU-607728	41
SKU-239162	36
SKU-665945	16
SKU-894238	16
SKU-646568	27
SKU-249154	34
SKU-952912	16
SKU-438927	19
SKU-676138	26
SKU-642988	19
SKU-784630	42



Cost	Weight (kg)	SKU	Dimensions
● Valide 100%	● Valide 100%	● Valide 100%	● Valide
● Erreur 0%	● Erreur 0%	● Erreur 0%	● Erreur
● Vide 0%	● Vide 0%	● Vide 0%	● Vide
71	6.23	SKU-619955	17x18x14 cm
132	3.46	SKU-367525	43x36x4 cm
1309	7.28	SKU-874528	45x31x8 cm
954	8.67	SKU-167410	43x12x16 cm
703	2.77	SKU-668353	22x31x2 cm
834	6.65	SKU-772806	15x39x17 cm
230	5.51	SKU-640119	22x11x7 cm
1159	2.94	SKU-154609	35x22x7 cm
135	3.59	SKU-791801	59x10x10 cm
926	7.17	SKU-129497	51x33x8 cm
1241	7.9	SKU-607728	25x22x19 cm
1294	0.41	SKU-239162	54x13x18 cm
1440	2.12	SKU-665945	14x34x16 cm
1424	3.55	SKU-894238	56x25x8 cm
951	9.88	SKU-646568	18x21x17 cm
694	8.76	SKU-249154	48x10x17 cm
1073	6.63	SKU-952912	15x36x14 cm
453	7.76	SKU-438927	49x35x12 cm
839	0.78	SKU-676138	22x32x19 cm
1039	9.34	SKU-642988	43x37x14 cm
235	0.36	SKU-784630	28x14x2 cm
1054	4.7	SKU-968755	60x17x4 cm
1082	8.23	SKU-170209	19x33x7 cm
611	3.01	SKU-409176	52x16x15 cm
201	7.91	SKU-783285	50x21x13 cm
1346	8.94	SKU-353870	19x29x6 cm

MERGE 2 QUERIES

1 ² ₃ Cost	1.2 Weight (kg)	A ^B _C SKU	A ^B _C Dimensions	A ^B _C SKU	1 ² ₃ souscatID
● Valide 100% ● Erreur 0% ● Vide 0%	● Valide 100% ● Erreur 0% ● Vide 0%	● Valide 100% ● Erreur 0% ● Vide 0%	● Valide 100% ● Erreur 0% ● Vide 0%	● Valide 100% ● Erreur 0% ● Vide 0%	● Valide 100% ● Erreur 0% ● Vide 0%
71	6.23	SKU-619955	17x18x14 cm	SKU-619955	5
132	3.46	SKU-367525	43x36x4 cm	SKU-367525	7
1309	7.28	SKU-874528	45x31x8 cm	SKU-874528	16
954	8.67	SKU-167410	43x12x16 cm	SKU-167410	18
703	2.77	SKU-668353	22x31x2 cm	SKU-668353	25
834	6.65	SKU-772806	15x39x17 cm	SKU-772806	17
230	5.51	SKU-640119	22x11x7 cm	SKU-640119	29
1159	2.94	SKU-154609	35x22x7 cm	SKU-154609	25
135	3.59	SKU-791801	59x10x10 cm	SKU-791801	35
926	7.17	SKU-129497	51x33x8 cm	SKU-129497	23
1241	7.9	SKU-607728	25x22x19 cm	SKU-607728	41
1294	0.41	SKU-239162	54x13x18 cm	SKU-239162	36
1440	2.12	SKU-665945	14x34x16 cm	SKU-665945	16
1424	3.55	SKU-894238	56x25x8 cm	SKU-894238	16
951	9.88	SKU-646568	18x21x17 cm	SKU-646568	27
694	8.76	SKU-249154	48x10x17 cm	SKU-249154	34
1073	6.63	SKU-952912	15x36x14 cm	SKU-952912	16
453	7.76	SKU-438927	49x35x12 cm	SKU-438927	19
839	0.78	SKU-676138	22x32x19 cm	SKU-676138	26
1039	9.34	SKU-642988	43x37x14 cm	SKU-642988	19
235	0.36	SKU-784630	28x14x2 cm	SKU-784630	42

ADD QUERIES

position	commandeID	clientID	produitID	NB vente	date commande	date livraison
1	20-0001	19	DRO-786	117	04.01.2020	11.01.2020
2	20-0002	46	ACC-700	24	10.01.2020	16.01.2020
3	20-0003	27	AUD-836	2	11.01.2020	16.01.2020
4	20-0004	46	AUD-703	116	13.01.2020	19.01.2020
5	20-0005	16	SMR-766	129	15.01.2020	18.01.2020
6	20-0006	5	DRO-714	101	17.01.2020	22.01.2020

6	20-0006		5	DRO-714	101	17.01.2021	22.01.2021		
7	20-0007	position		commandeID	clientID	produitID	NB vente	date commande	date livraison
8	20-0008		1	21-0250	35	DRO-691	135	03.01.2021	06.01.2021
9	20-0009		2	21-0251	99	DRO-652	18	05.01.2021	12.01.2021
10	20-0010		3	21-0252	28	NET-343	21	08.01.2021	15.01.2021
11	20-0011		4	21-0253	90	SMR-668	139	09.01.2021	13.01.2021
12	20-0012		5	21-0254	83	NET-835	52	10.01.2021	13.01.2021
13	20-0013		6	21-0255	23	AUD-758	58	10.01.2021	15.01.2021

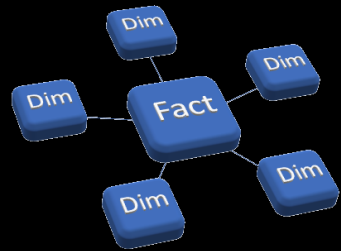
position	commandeID	clientID	produitID	NB vente	date commande	date livraison
1	22-0001	19	COM-384	31	08.01.2022	11.01.2022
2	22-0002	85	DRO-311	63	09.01.2022	14.01.2022
3	22-0003	31	SMR-317	57	10.01.2022	15.01.2022
4	22-0004	51	HAP-900	106	11.01.2022	14.01.2022
5	22-0005	32	NET-431	55	11.01.2022	16.01.2022
6	22-0006	61	SMR-887	41	11.01.2022	18.01.2022
7	22-0007	27	HAP-320	9	14.01.2022	17.01.2022
8	22-0008	31	SMR-342	113	15.01.2022	20.01.2022
9	22-0009	6	CAM-828	122	15.01.2022	19.01.2022
10	22-0010	92	WER-988	28	17.01.2022	20.01.2022
11	22-0011	81	DRO-835	6	18.01.2022	24.01.2022
12	22-0012	91	CAM-193	46	19.01.2022	26.01.2022

ADD QUERIES

commandeID	clientID	produitID	NB vente	date commande	date livraison
20-0001	19	DRO-786	117	04.01.2020	11.01.2020
20-0002	46	ACC-700	24	10.01.2020	16.01.2020
20-0003	27	AUD-836	2	11.01.2020	16.01.2020
20-0004	46	AUD-703	116	13.01.2020	19.01.2020
20-0005	16	SMR-766	129	15.01.2020	18.01.2020
20-0006	5	DRO-714	101	17.01.2020	22.01.2020
20-0007	36	SMR-672	41	19.01.2020	25.01.2020
20-0008	83	NET-279	31	19.01.2020	22.01.2020
20-0009	27	CAM-628	50	19.01.2020	24.01.2020
20-0010	18	COM-384	45	19.01.2020	25.01.2020
21-0254	83	NET-835	52	10.01.2021	13.01.2021
21-0255	23	AUD-758	58	10.01.2021	15.01.2021
21-0256	93	ACC-723	88	11.01.2021	17.01.2021
21-0257	7	ACC-997	96	12.01.2021	16.01.2021
21-0258	14	DRO-685	54	14.01.2021	20.01.2021
21-0259	79	AUD-871	146	17.01.2021	22.01.2021
21-0260	77	AUD-964	59	20.01.2021	26.01.2021
21-0261	90	AUD-964	52	20.01.2021	25.01.2021
22-0003	31	SMR-317	57	10.01.2022	15.01.2022
22-0004	51	HAP-900	106	11.01.2022	14.01.2022
22-0005	32	NET-431	55	11.01.2022	16.01.2022
22-0006	61	SMR-887	41	11.01.2022	18.01.2022
22-0007	27	HAP-320	9	14.01.2022	17.01.2022
22-0008	31	SMR-342	113	15.01.2022	20.01.2022
22-0009	6	CAM-828	122	15.01.2022	19.01.2022
22-0010	92	WER-988	28	17.01.2022	20.01.2022
22-0011	81	DRO-835	6	18.01.2022	24.01.2022
22-0012	91	CAM-193	46	19.01.2022	26.01.2022

Best Practices in POWER query

- Give clear and understandable names to your queries and steps
- Give a meaningful name to your columns as they will be part of your model
- Verify the data types in each column before importing
- Get rid of unused columns
- Use preferably the import method (instead of direct method)
- Do not load queries/tables that you do not need in your model
- Activate the formula bar



DATA MODELING

Relational database Concepts



Power BI
Desktop

RELATIONAL DATABASE

- Stores and provides access to data points that are related to one another.
- Based on the relational model, an intuitive, straightforward way of representing data in tables.
- Each row in the table is a record with a unique ID called the key.
- The columns of the table hold attributes of the data.
- The goal is to make it easy to establish relationships among data points in order to quickly and efficiently summarize data from the various linked entities in the data model.

Key concepts

Tables (Relations):

Data is organized into tables, with rows representing records and columns representing attributes.

Keys:

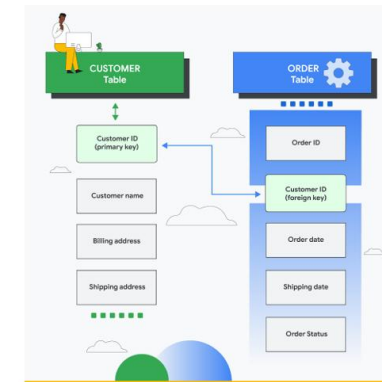
A **primary key** is a column that can reliably identify each individual record in a table. It can be an existing column (or combination of) from the table (a natural key) or from a new attribute containing a unique ID created for this purpose (a surrogate key).

A **foreign key** is a column in a table that refers to the primary key of another table, linking these two tables. It usually does not have a unique value.

Unique identifiers (primary keys) and relationships between tables (foreign keys) ensure data integrity and allow for efficient querying.

Relationships:

Relationships determine dependency between a primary key in one table and a foreign key in another table..



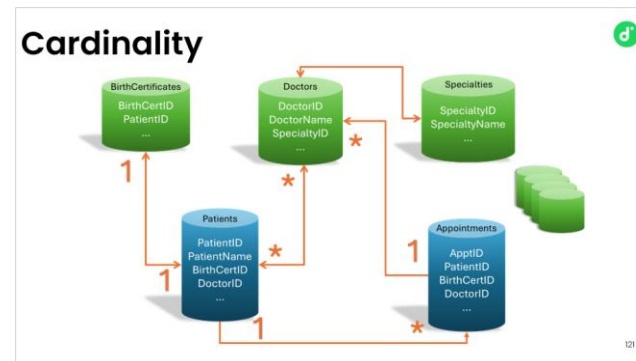
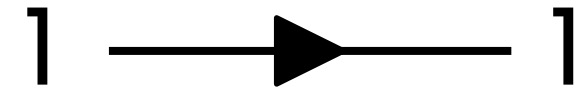
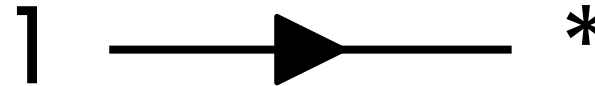
Cardinality

- The cardinality is the numerical relationship between rows of one table and rows in another. Common cardinalities include:

A many-to-many relationship.

A one-to-many relationship

A one-to-one relationship



It is recommended to avoid many-to-many relationships but to transform them through the creation of joining tables.

Fact & dimension tables

Fact tables

- Contains numeric data that will be aggregated in the visualizations. (i.e. fctSales, fctProfits). It needs to include the keys to the related dimensions tables.

Dimension tables

- Contains descriptive info that can be used to describe, group, categorize or filter the fact tables (i.e. dimCustomers, dimProducts, etc)
- It requires a key field (Primary key)

* Best practice: do not mix fact and dimension data in the same table (optimize the performance)

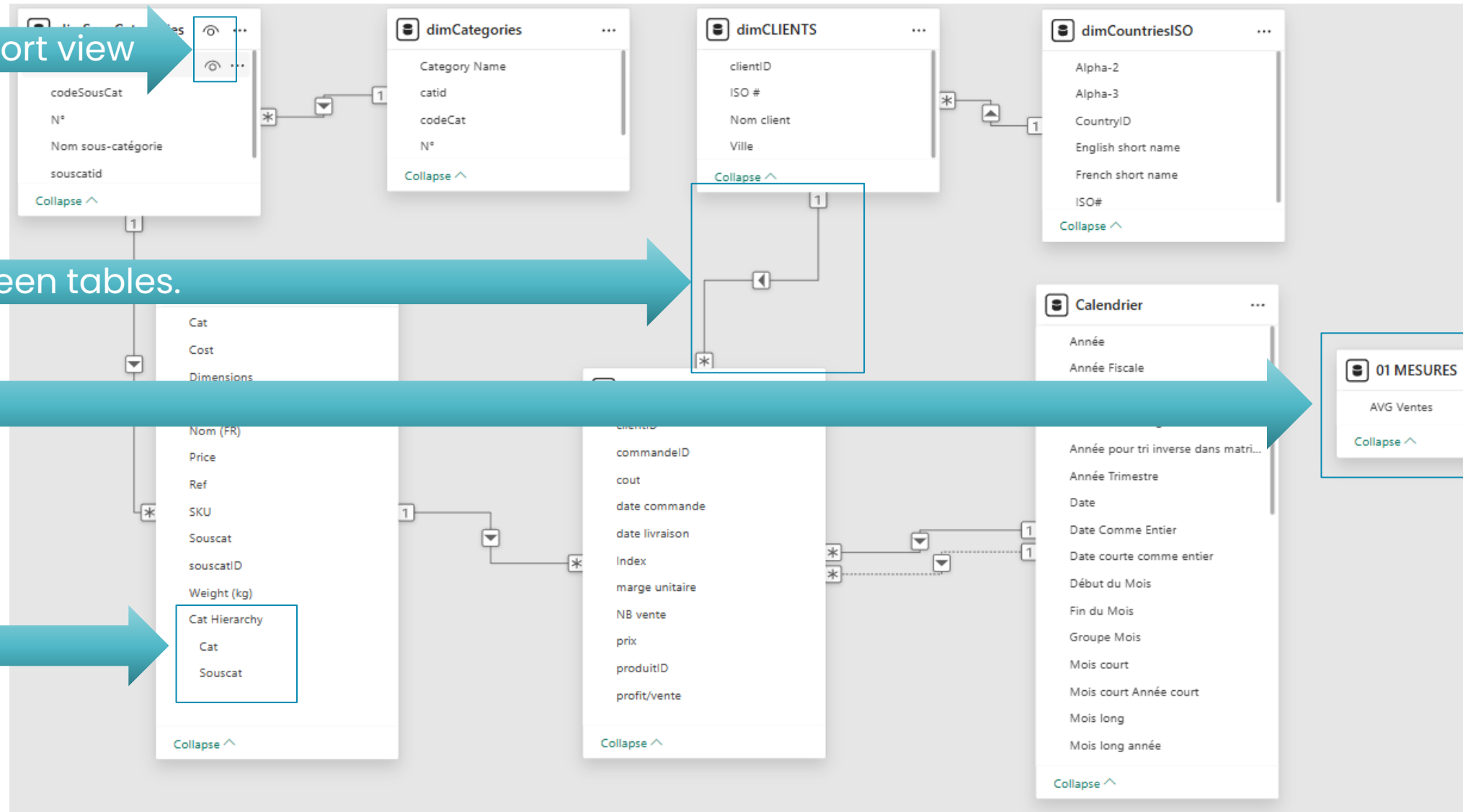
The data model

Visibility in the report view

Relationships between tables.

Organize Measures

Create hierarchies



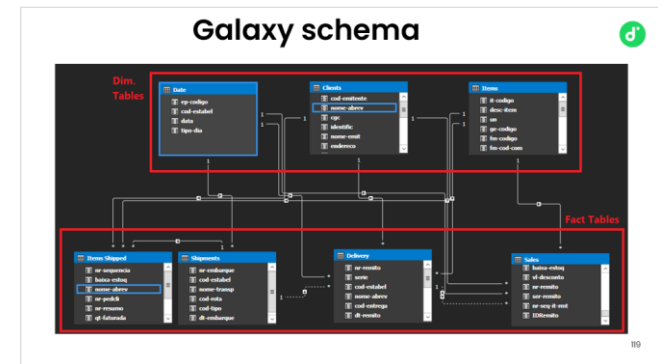
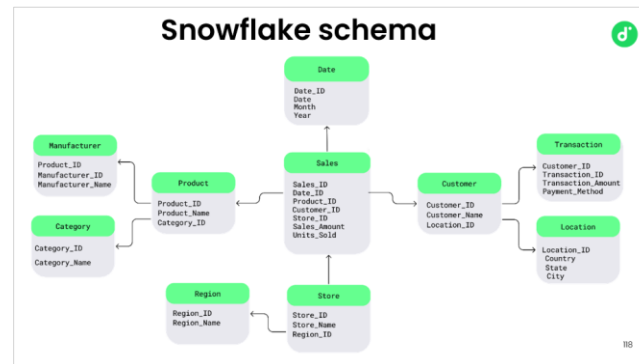
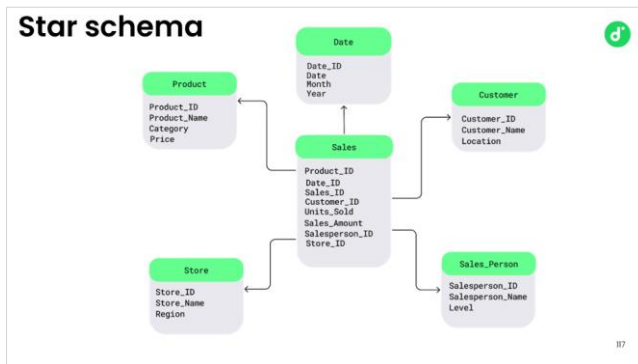
Types of Data Models

- The representation of tables with their attributes (columns) and relationships is called the physical data model.

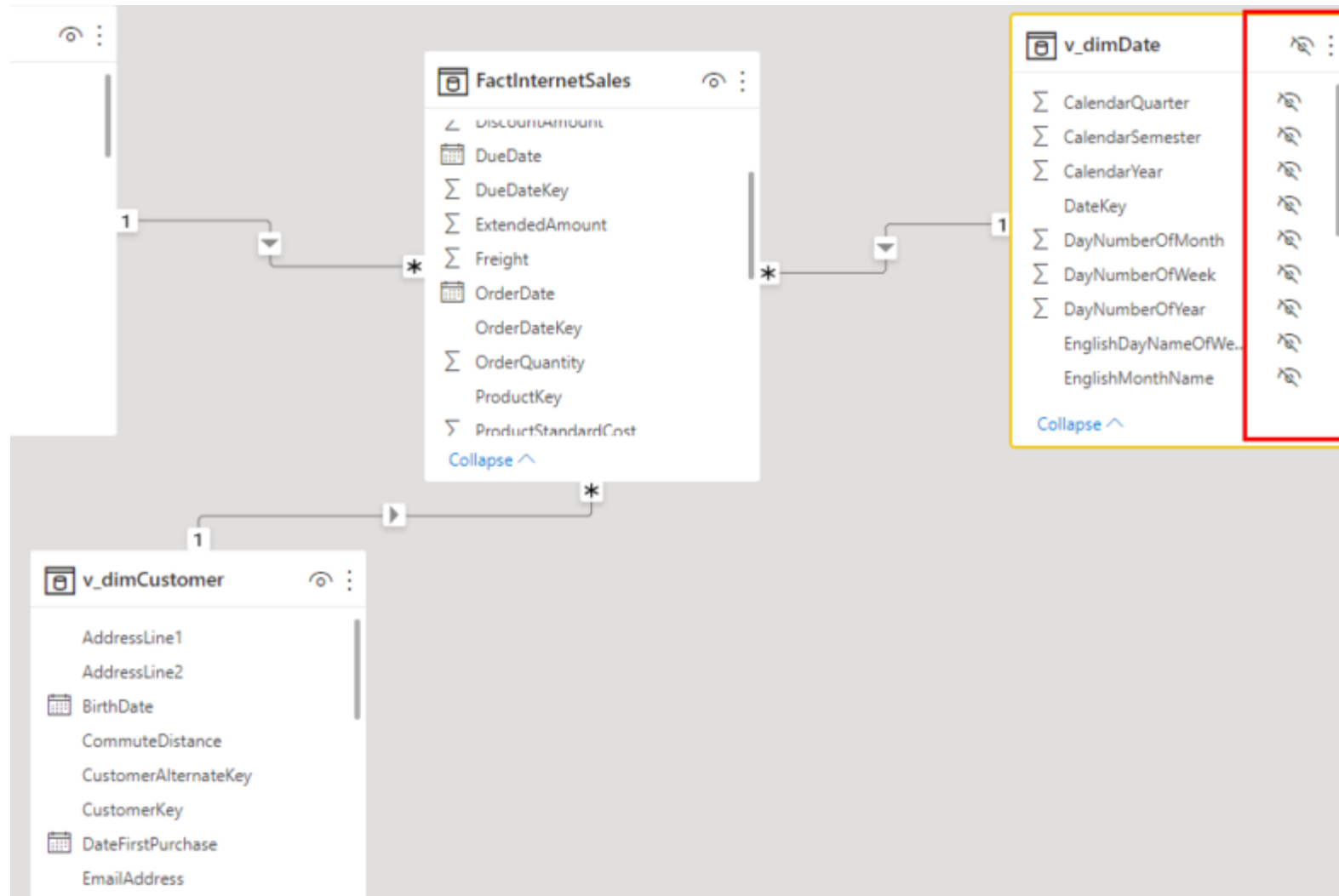
When every dimension table is linked directly to the fact table, we talk about star schema.

When some tables are connected to dimension tables, not directly to the fact table, this is a snowflake schema.

When several fact and dimension tables coexist in the data model, it is described as a galaxy schema (or fact constellation schema)

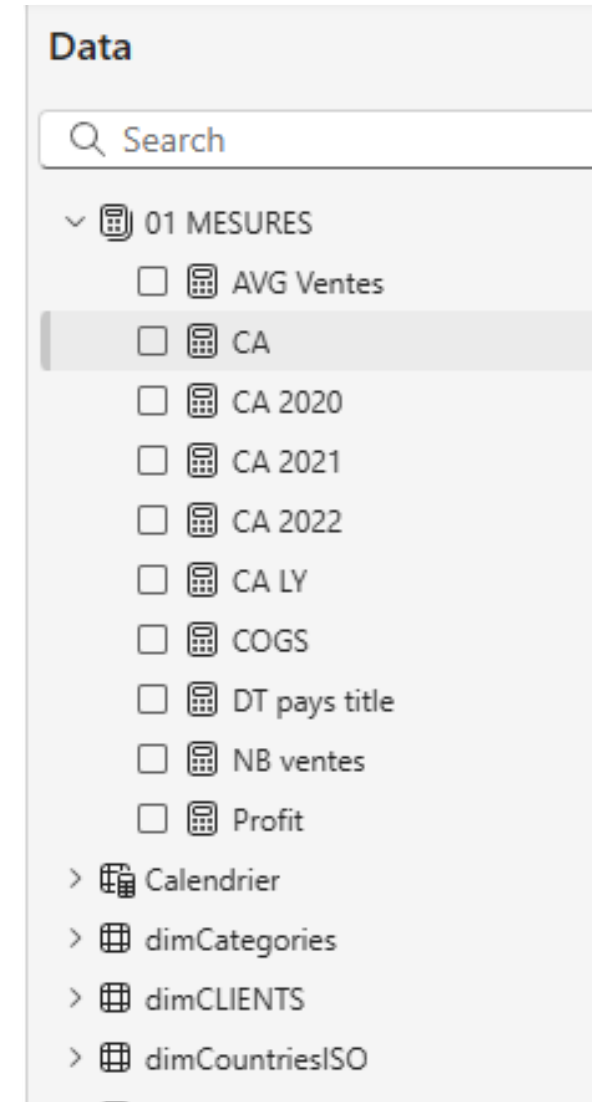


Hide fields or tables from the report view

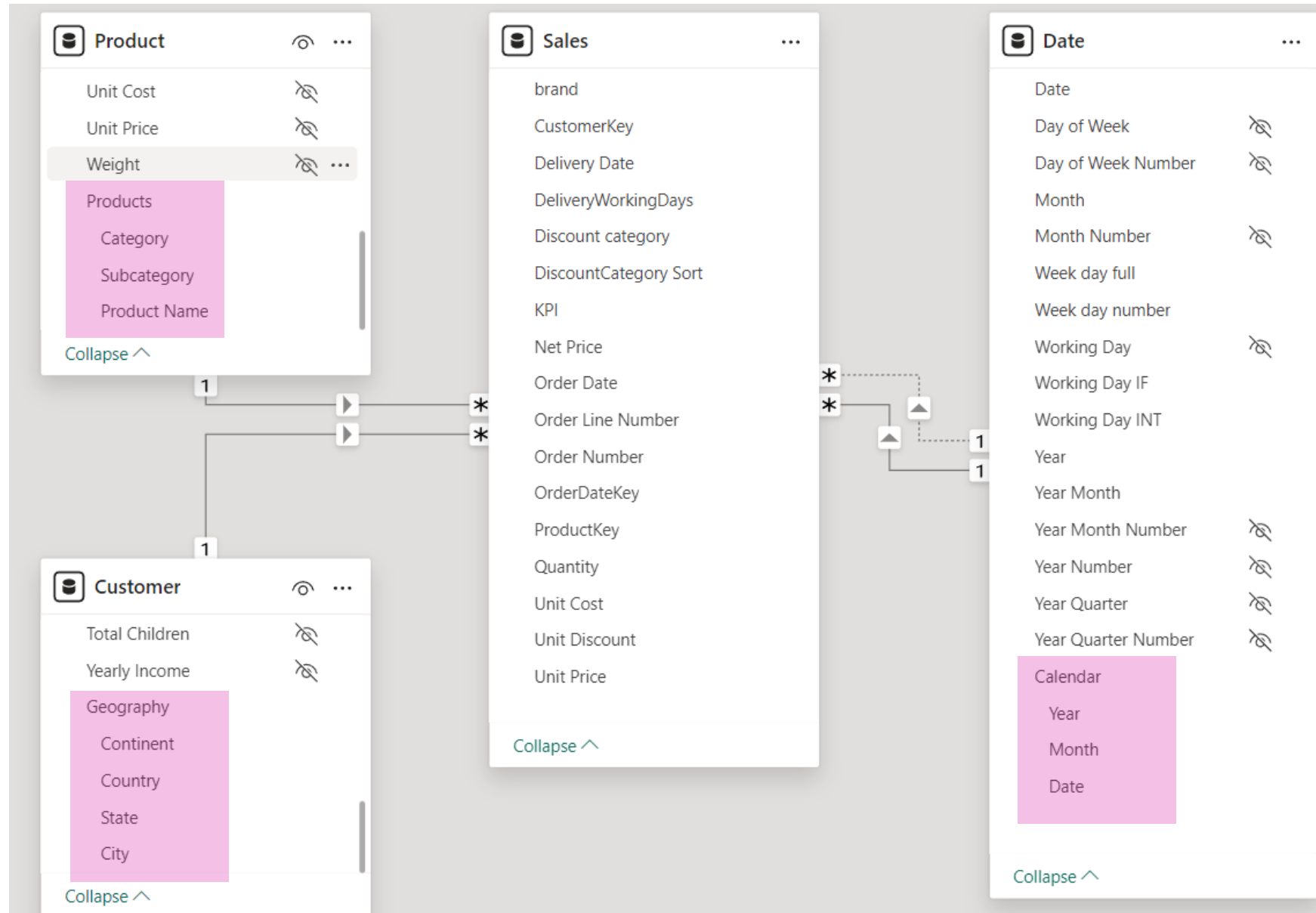


Organizing measures

- Name measures explicitly with a clear name
- Relocate all measures in one table
- Regroupe within folders \subfolders; otherfolder
- Rename folders for order of appearance



Create hierarchies



Best Practices in model view

- Always use a separate date table marked as such for time intelligence
- Whenever possible, use the star schema and identify the dim and facts tables
- Verify the relationships between tables
- Stick to the data normalization , i.e. use YTD, LY, YoY
- Hide the columns that are not meaningful to the user
- Place your DAX measures in a dedicated table (if needed hide the original one)
- Use DIVIDE() (treats division by 0) instead of /



DAX

Data Analysis EXpressions



Power BI

Desktop



DAX is the formula language in Power BI

It can be used for creating:

- Calculated tables
- Calculated columns
- Measures

Calculated table: DAX Calendar

Calendrier =

VAR StartDate = DATE(2020, 1, 1)

VAR EndDate = DATE(2021, 12, 31)

VAR DateList = CALENDAR(StartDate, EndDate)

RETURN

ADDCOLUMNS (

CALENDAR (DATE (2020, 1, 1), DATE (2022, 12, 31)),

"Date Comme Entier", FORMAT ([Date], "YYYYMMDD"),

"Année", YEAR ([Date]),

"Année pour tri inverse dans matrices", YEAR ([Date]) *-1,

"Numéro Mois", FORMAT ([Date], "MM"),

"Numéro Année Mois", FORMAT ([Date], "YYYY/MM"),

"Semestre", "S" & IF (MONTH ([Date]) <= 6, 1, 2),

"Année Mois court", FORMAT ([Date], "YYYY/mmm"),

"Année Mois long", FORMAT ([Date], "YYYY mmmm"),

"Mois court", FORMAT ([Date], "mmm"),

"Mois long", FORMAT ([Date], "mmmm")

"Mois long année", FORMAT ([Date], "mmmm YYYY"),

"Mois court Année court", FORMAT ([Date], "MM-yy"),

...

)

The screenshot shows a Power BI Desktop window with a table named 'Calendar'. The table contains the following columns:

- Date
- Year
- Month
- Year for matrix sorting (Year * -1)
- Month Number
- Year-Month Number
- Semester
- Short Year-Month
- Long Year-Month
- Short Month
- Long Month
- Long Month-Year
- Short Month Short Year

The table is displayed in a grid view with a date range from 2020-01-01 to 2021-12-31.

125

Calculated column

$$\text{Profit/vente} = \text{FactCommandes}[\text{marge unitaire}] * \text{FactCommandes}[\text{NB vente}]$$

001.MPBI01-SwiCrossoft • Last saved: 1/31/2025 at 11:05 PM

Search

Ludovic MS Business account

File Home Help External tools Table tools Column tools

Name profit/vente 123 Whole number \$% General \$ % , .00 Auto Σ Sum

1 profit/vente = FactCommandes[marge unitaire] * FactCommandes[NB vente]

Index	commandeID	clientID	produitID	NB vente	prix	cout	marge unitaire	profit/vente	date commande	date livraison
1	20-0001	19	DRO-786	117	1 100.00 CHF	725 CHF	375	43875	samedi, 4 janvier 2020	samedi, 11 janvier 2020
2	20-0002	46	ACC-700	24	165.00 CHF	100 CHF	65	1560	vendredi, 10 janvier 2020	jeudi, 16 janvier 2020
3	20-0003	27	AUD-836	2	1 579.00 CHF	897 CHF	682	1364	samedi, 11 janvier 2020	jeudi, 16 janvier 2020
4	20-0004	46	AUD-703	116	462.00 CHF	326 CHF	136	15776	lundi, 13 janvier 2020	dimanche, 19 janvier 2020
5	20-0005	16	SMR-766	129	1 860.00 CHF	1 484 CHF	376	48504	mercredi, 15 janvier 2020	samedi, 18 janvier 2020
6	20-0006	5	DRO-714	101	1 836.00 CHF	1 500 CHF	336	33936	vendredi, 17 janvier 2020	mercredi, 22 janvier 2020
7	20-0007	36	SMR-672	41	1 191.00 CHF	925 CHF	266	10906	dimanche, 19 janvier 2020	samedi, 25 janvier 2020
8	20-0008	83	NET-279	31	134.00 CHF	101 CHF	33	1023	dimanche, 19 janvier 2020	mercredi, 22 janvier 2020
9	20-0009	27	CAM-628	50	528.00 CHF	309 CHF	219	10950	dimanche, 19 janvier 2020	vendredi, 24 janvier 2020
10	20-0010	18	COM-384	45	297.00 CHF	179 CHF	118	5310	dimanche, 19 janvier 2020	samedi, 25 janvier 2020
11	20-0011	53	ACC-948	137	1 704.00 CHF	1 365 CHF	339	46443	dimanche, 19 janvier 2020	samedi, 25 janvier 2020
12	20-0012	80	CAM-673	3	1 458.00 CHF	1 053 CHF	405	1215	mardi, 21 janvier 2020	vendredi, 24 janvier 2020
13	20-0013	64	COM-912	133	1 601.00 CHF	922 CHF	679	90307	mardi, 21 janvier 2020	lundi, 27 janvier 2020
14	20-0014	83	AUD-662	95	2 244.00 CHF	1 372 CHF	872	82840	dimanche, 26 janvier 2020	mercredi, 29 janvier 2020
15	20-0015	72	SMR-668	61	1 995.00 CHF	1 460 CHF	535	32635	lundi, 27 janvier 2020	samedi, 1 février 2020
16	20-0016	71	AUD-468	139	1 513.00 CHF	1 188 CHF	325	45175	mardi, 28 janvier 2020	vendredi, 31 janvier 2020
17	20-0017	7	AUD-396	81	675.00 CHF	487 CHF	188	15228	jeudi, 30 janvier 2020	jeudi, 6 février 2020
18	20-0018	36	CAM-304	54	1 780.00 CHF	1 398 CHF	382	20628	jeudi, 30 janvier 2020	mardi, 4 février 2020
19	20-0019	48	ACC-926	128	423.00 CHF	285 CHF	138	17664	jeudi, 30 janvier 2020	jeudi, 6 février 2020
20	20-0020	22	WER-312	5	754.00 CHF	435 CHF	319	1595	dimanche, 2 février 2020	vendredi, 7 février 2020

Data

Search

- 01 MESURES
- Calendrier
- dimCategories
- dimCLIENTS
- dimCountriesISO
- dimProduits
- dimSousCategories
- FactCommandes
 - clientID
 - commandeID
 - cout
 - date commande
 - date livraison
 - Index
 - marge unitaire
 - NB vente
 - prix

DAX MEASURES

IMPLICIT MEASURES

Created automatically by Power BI. Dragging a field into a visualization well will create an implicit measure (automatic sum, count, etc)

EXPLICIT MEASURES

Created by explicitly typing a DAX code after hitting the 'New Measure' button.

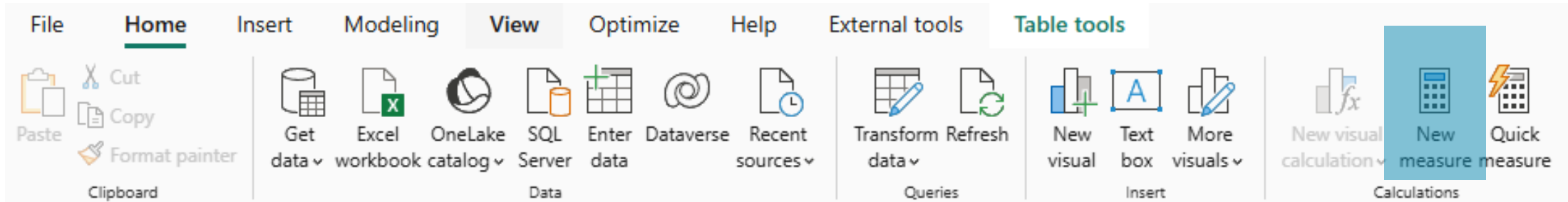
QUICK MEASURES

List of predefined DAX measures proposed by Power BI. No code needed.

The screenshot displays the Power BI interface. On the left, a data table is visible with columns: Cat, CA, Sum of Total sale, and CA average per Subcategory Name. The table lists various product categories like Caméra, Jeux vidéo, Ordinateur, and Smartphone, along with their respective sales figures. On the right, the 'Quick measure' pane is open, showing a list of predefined measures. The 'Base value' is set to 'CA' and the 'Category' is set to 'Subcategory Name'. The 'Visualizations' pane on the far right shows a bar chart visualization. A legend at the bottom of the Quick Measure pane identifies the three types of measures: Explicit Measure (CA), Implicit Measure (Sum of Total sale), and Quick Measure (CA average per Subcategory Name).

Cat	CA	Sum of Total sale	CA average per Subcategory Name
Caméra	6790551 CHF	6790551 CHF	1358110 CHF
Jeux vidéo	4121711 CHF	4121711 CHF	824342 CHF
Ordinateur	7311923 CHF	7311923 CHF	1462265 CHF
Mini PC	2319817 CHF	2319817 CHF	2319817 CHF
Ordinateur de bureau	828246 CHF	828246 CHF	828246 CHF
Ordinateur portable	1567650 CHF	1567650 CHF	1567650 CHF
Tablette	2155386 CHF	2155386 CHF	2155386 CHF
Tout-en-un	439714 CHF	439714 CHF	439714 CHF
Smartphone	7362838 CHF	7362838 CHF	1472568 CHF
Premier prix	1137207 CHF	1137207 CHF	1137207 CHF
Smartphone 2"	1997691 CHF	1997691 CHF	1997691 CHF
Smartphone 3"	1831062 CHF	1831062 CHF	1831062 CHF
Téléphone de jeu	1031868 CHF	1031868 CHF	1031868 CHF
Téléphone pliable	1365010 CHF	1365010 CHF	1365010 CHF
Total	25586123 CHF	25586123 CHF	1279306 CHF

EXPLICIT DAX MEASURES



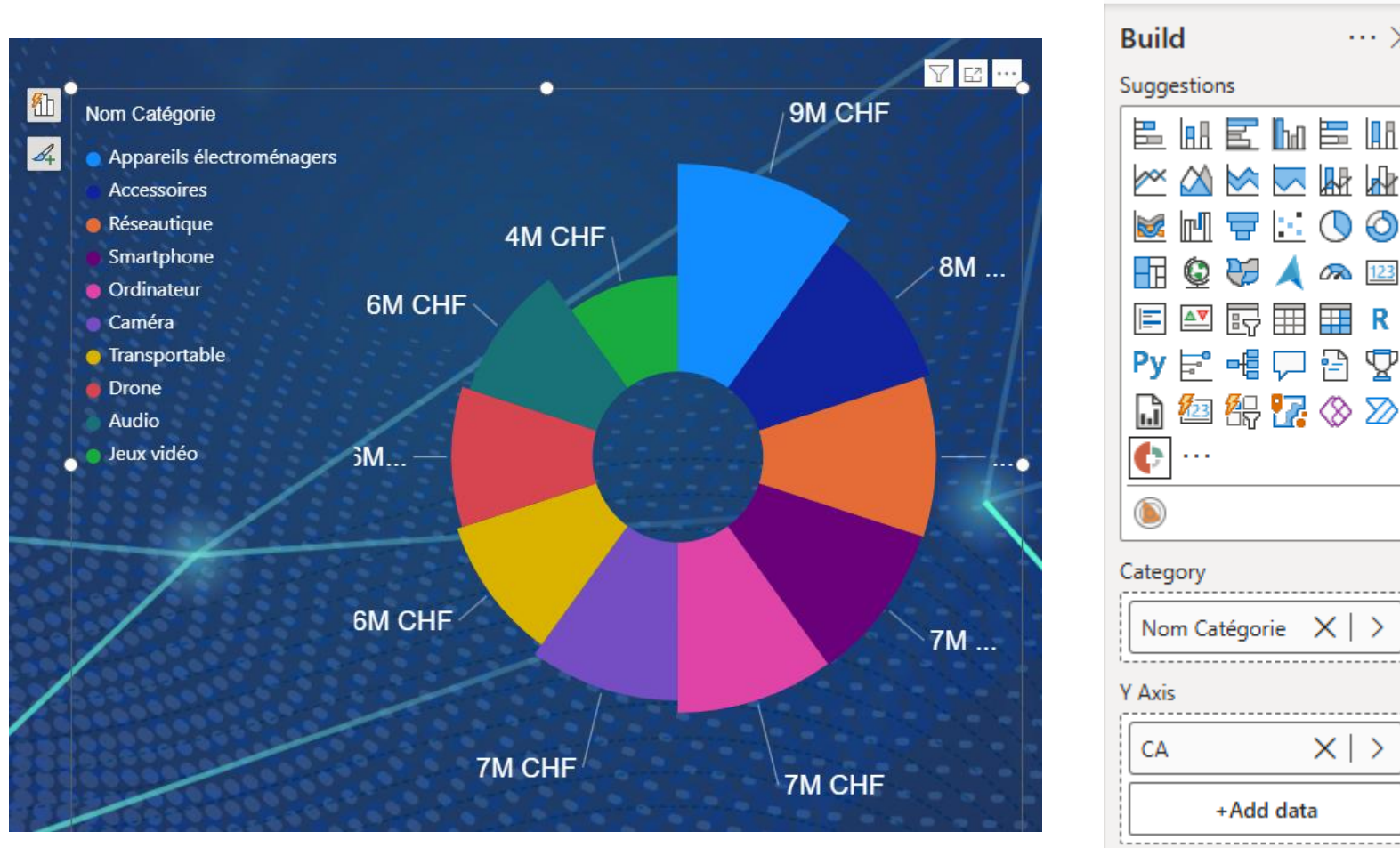
TotalSales = SUM(Sales[Amount])

TotalProducts = COUNTROWS(dimProducts[ProductID])

FilteredSales = CALCULATE(
 SUM(Sales[Amount]),
 FILTER(Sales,
 Sales[Region] = "North"))

Example of use of a DAX measure

$CA = \text{SUMX}(\text{FactCommandes}, \text{FactCommandes}[\text{NB vente}] * \text{RELATED}(\text{dimProduits}[\text{Price}]))$



DAX CALCULATED COLUMN

The screenshot shows the Microsoft Power BI Desktop interface with the 'Column tools' ribbon selected. The ribbon contains several groups of options:

- Name:** 'Total sale'
- Data type:** 'Whole number'
- Format:** 'Currency' (with a dropdown menu)
- Summarization:** 'Sum' (with a dropdown menu)
- Data category:** 'Uncategorized' (with a dropdown menu)
- Sort:** 'Sort by column' (with a dropdown menu)
- Groups:** 'Data groups' (with a dropdown menu)
- Relationships:** 'Manage relationships' (with a dropdown menu)
- Calculations:** 'New column Calculations' (highlighted in blue)

Below the ribbon, the formula bar shows the calculated column formula:

```
1 Total sale = [NB vente] * FactCommandes[prix]
```

The data table below the formula bar shows the following columns and values:

Index	commandeID	clientID	produitID	NB vente	prix	cout	marge unitaire	profit/vente	date commande	date livraison	Total sale
1	20-0001	19	DRO-786	117	1 100 CHF	725 CHF	375	43875	samedi, 4 janvier 2020	samedi, 11 janvier 2020	128 700 CHF
2	20-0002	46	ACC-700	24	165 CHF	100 CHF	65	1560	vendredi, 10 janvier 2020	jeudi, 16 janvier 2020	3 960 CHF
3	20-0003	27	AUD-836	2	1 579 CHF	897 CHF	682	1364	samedi, 11 janvier 2020	jeudi, 16 janvier 2020	3 158 CHF
4	20-0004	46	AUD-703	116	462 CHF	326 CHF	136	15776	lundi, 13 janvier 2020	dimanche, 19 janvier 2020	53 592 CHF
5	20-0005	16	SMR-766	129	1 860 CHF	1 484 CHF	376	48504	mercredi, 15 janvier 2020	samedi, 18 janvier 2020	239 940 CHF
6	20-0006	5	DRO-714	101	1 836 CHF	1 500 CHF	336	33936	vendredi, 17 janvier 2020	mercredi, 22 janvier 2020	185 436 CHF
7	20-0007	36	SMR-672	41	1 191 CHF	925 CHF	266	10906	dimanche, 19 janvier 2020	samedi, 25 janvier 2020	48 831 CHF

`prix = RELATED(dimProducts[Price])`

`Total Sale = FactCommandes[NB vente] * FactCommandes[Prix]`

\$120,17M

Total Sales

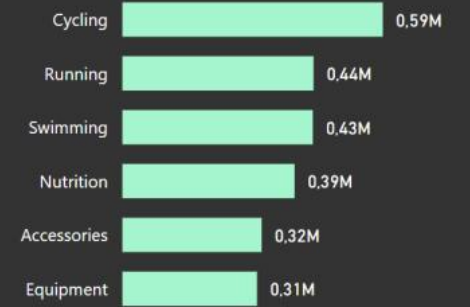


Sales



2M

Total Units Sold



POWER VIEW Visuals in Power BI



Power BI

Desktop

Visualizations pane

The image shows the 'Visualizations' pane in a data analytics tool. It features a 'Build visual' section with a grid of icons for different chart types. Below this is a 'Values' section with a text input field 'Add data fields here'. At the bottom, there are settings for 'Drill through' (Cross-report: Off, Keep all filters: On) and another text input field 'Add drill-through fields here'. Callout boxes point to specific icons in the grid: 'Build visual' points to the top-left icon; 'Stacked chart row' points to a stacked bar chart icon; 'Line, area & combo chart row' points to a line chart icon; 'KPI Visualization' points to a KPI card icon; 'Slicer' points to a vertical bar chart icon; 'Format your visual' points to the top-right icon; 'Pie + donut chart' points to a pie chart icon; 'Card and multi-row card' points to a card icon; and 'Table, Matrix' points to a table/matrix icon.

Build visual

Stacked chart row

Line, area & combo chart row

KPI Visualization

Slicer

Format your visual

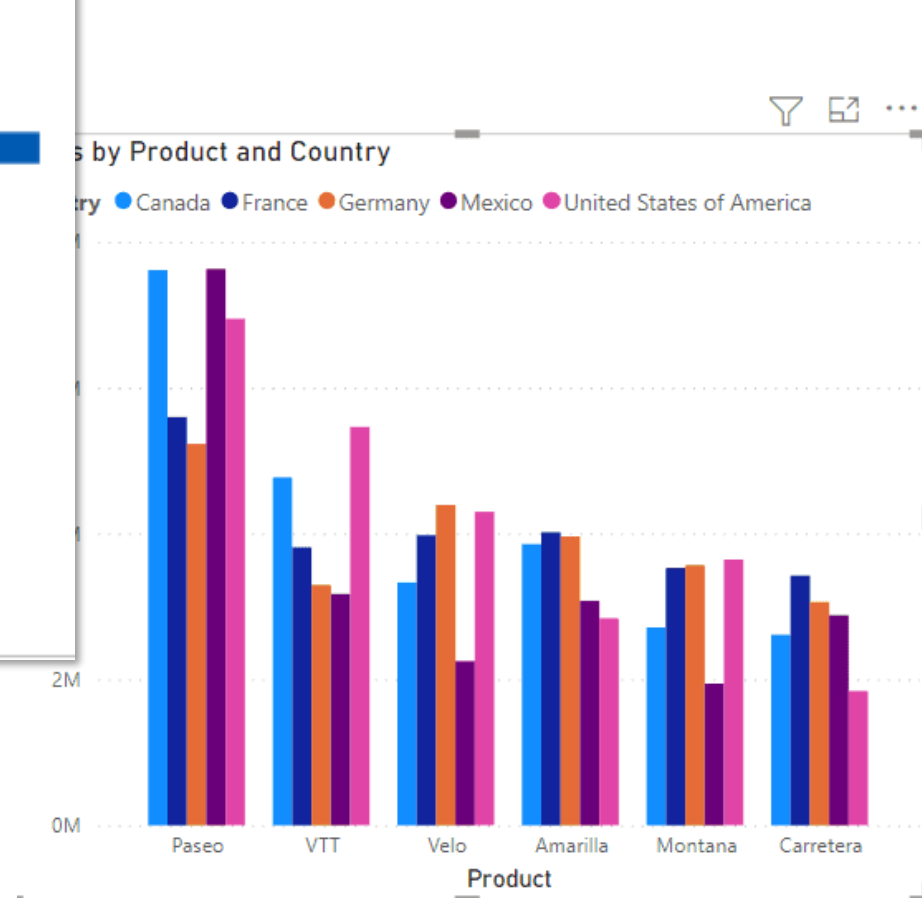
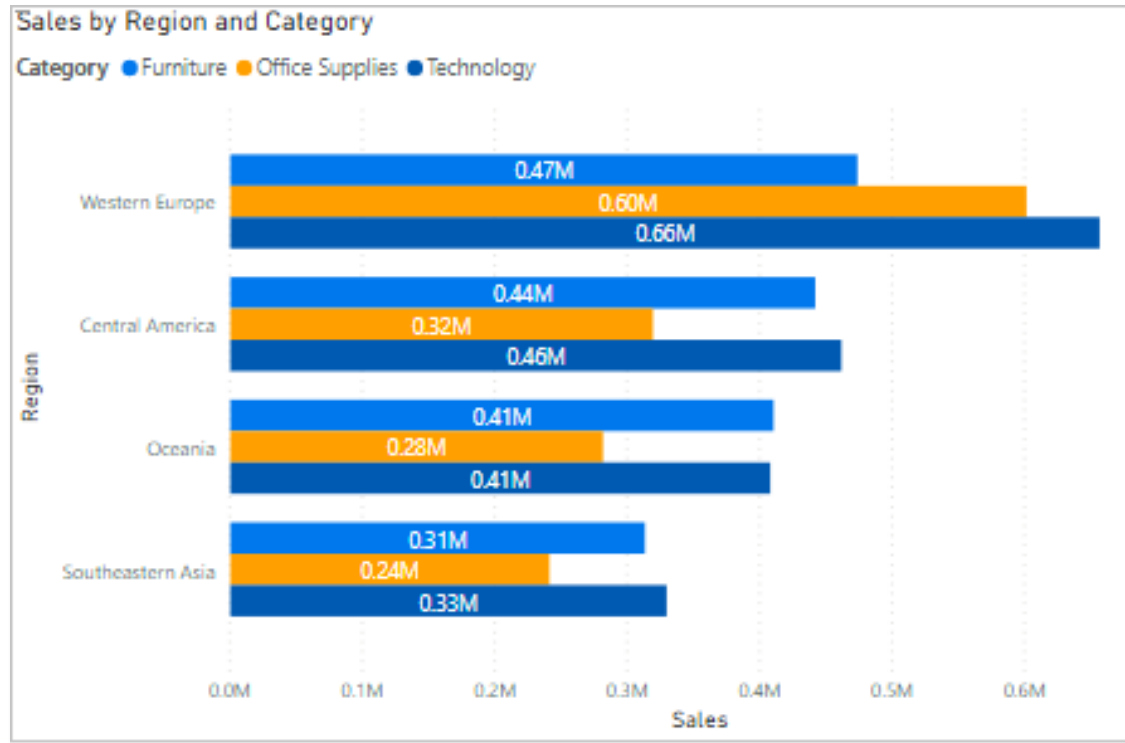
Pie + donut chart

Card and multi-row card

Table, Matrix

Clustered column & bar chart

when you want to compare values across different product categories, regions or time periods.



Filters

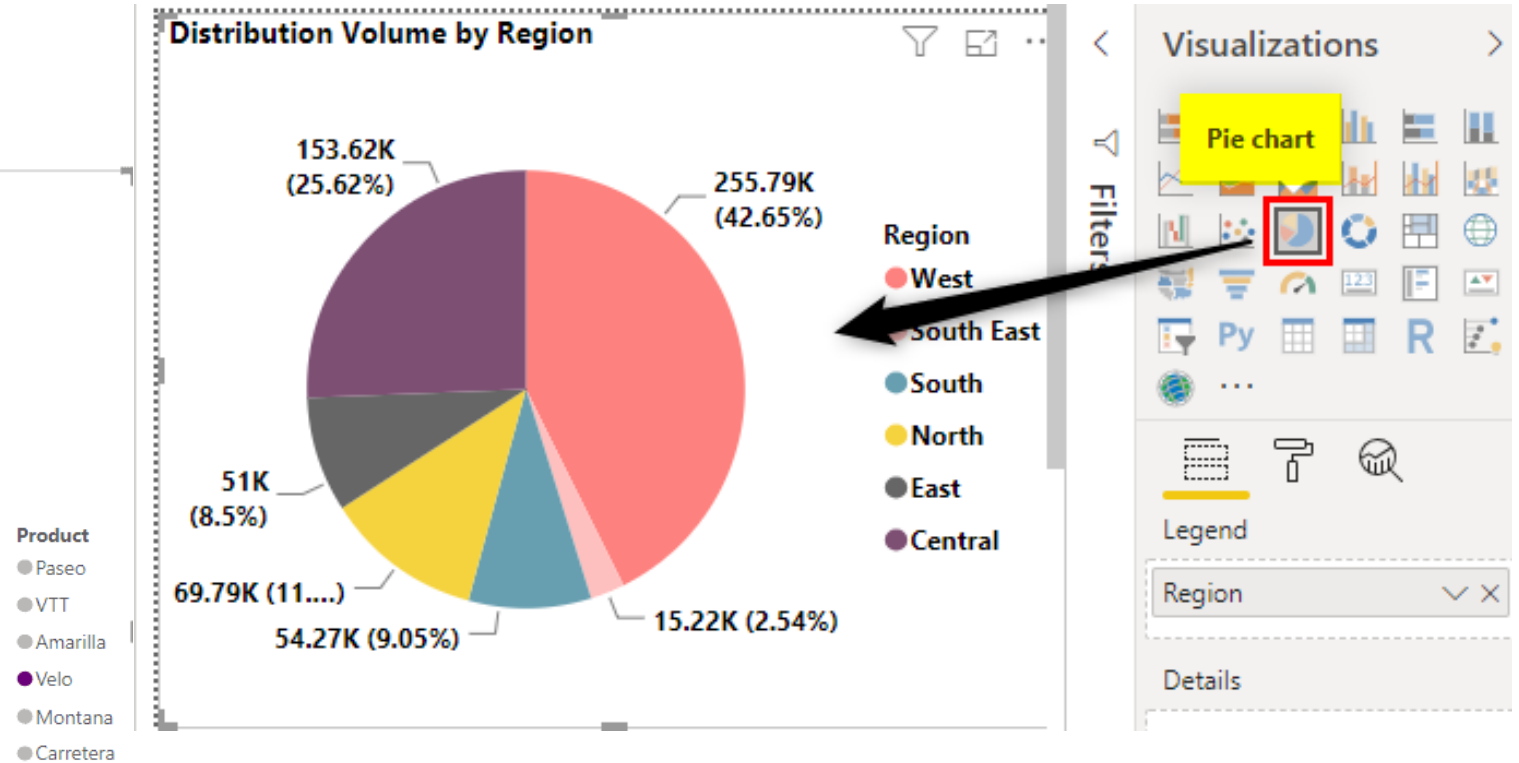
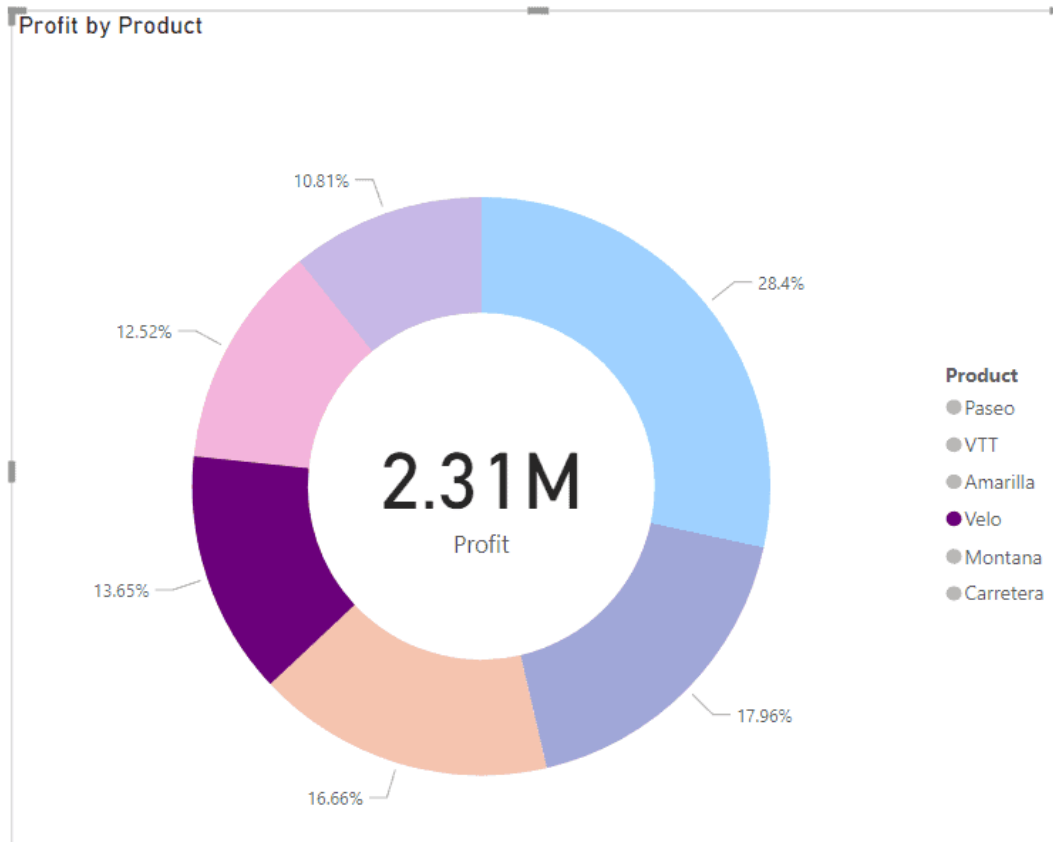
The configuration panel shows various chart types in a grid. A red box highlights the 'Clustered Column' chart type. Below the grid, the configuration settings are listed:

- Axis: Product
- Legend: Country
- Values: Sales
- Small multiples: (empty)
- Add data fields here: (empty)

Pie Charts

Illustrates the relative proportion or distribution of a variable

- Best use: 5-7 slices



Visualization : table

date commande	date livraison	commandeID	Nom client	Sum of NB vente	Sum of prix	CA		
samedi 4 janvier 2020	samedi 11 janvier 2020	20-0001	Horizon Vert	117	1'100.00 CHF	128'700		
vendredi 10 janvier 2020	jeudi 16 janvier 2020	20-0002	GrandTech	24	165.00 CHF	3'960		
samedi 11 janvier 2020	jeudi 16 janvier 2020	20-0003	TransAlp Logistique	2	1'579.00 CHF	3'158		
lundi 13 janvier 2020	dimanche 19 janvier 2020	20-0004	GrandTech	116	462.00 CHF	53'592		
mercredi 15 janvier 2020	samedi 18 janvier 2020	20-0005	SolarGen Europe	129	1'860.00 CHF	239'940		
vendredi 17 janvier 2020	mercredi 22 janvier 2020	20-0006	HauteRive Consulting	101	1'836.00 CHF	185'436		
dimanche 19 janvier 2020	mercredi 22 janvier 2020	20-0008	SummitPeak Technologies	31	134.00 CHF	4'154		
dimanche 19 janvier 2020	vendredi 24 janvier 2020	20-0009	TransAlp Logistique	50	528.00 CHF	26'400		
dimanche 19 janvier 2020	samedi 25 janvier 2020	20-0007	DataPoint Inc.	41	1'191.00 CHF	48'831		
dimanche 19 janvier 2020	samedi 25 janvier 2020	20-0010	MétoTech	45	297.00 CHF	13'365		
dimanche 19 janvier 2020	samedi 25 janvier 2020	20-0011	BluRiver Architecture	137	1'704.00 CHF	233'448		
mardi 21 janvier 2020	vendredi 24 janvier 2020	20-0012	AlpineWave	3	1'458.00 CHF	4'374		
mardi 21 janvier 2020	lundi 27 janvier 2020	20-0013	HighTech Sàrl	133	1'601.00 CHF	212'933		
dimanche 26 janvier 2020	mercredi 29 janvier 2020	20-0014	SummitPeak Technologies	95	2'244.00 CHF	213'180		
lundi 27 janvier 2020	samedi 1 février 2020	20-0015	AlpenScience AG	61	1'995.00 CHF	121'695		
mardi 28 janvier 2020	vendredi 31 janvier 2020	20-0016	EuroFuture Corp.	139	1'513.00 CHF	210'307		
jeudi 30 janvier 2020	mardi 4 février 2020	20-0018	DataPoint Inc.	54	1'780.00 CHF	96'120		
jeudi 30 janvier 2020	jeudi 6 février 2020	20-0017	Swiss Health Solutions	81	675.00 CHF	54'675		
jeudi 30 janvier 2020	jeudi 6 février 2020	20-0019	BaselLab Solutions	128	423.00 CHF	54'144		
dimanche 2 février 2020	vendredi 7 février 2020	20-0020	Horizon Durable	5	754.00 CHF	3'770		
dimanche 2 février 2020	samedi 8 février 2020	20-0021	NovaSuisse Technologies	30	577.00 CHF	17'310		
mardi 4 février 2020	dimanche 9 février 2020	20-0022	Swiss Finances SA	104	529.00 CHF	55'016		
vendredi 7 février 2020	mardi 11 février 2020	20-0023	HauteRive Consulting	60	577.00 CHF	34'620		
samedi 8 février 2020	jeudi 13 février 2020	20-0025	VisionDigit	79	834.00 CHF	65'886		
samedi 8 février 2020	samedi 15 février 2020	20-0024	AlphaVision Sàrl	29	661.00 CHF	19'169		
dimanche 9 février 2020	dimanche 16 février 2020	20-0026	PeakIT Solutions	89	273.00 CHF	24'297		

Build

Suggestions

Columns

date commande × | >

date livraison × | >

commandeID × | >

Nom client × | >

Sum of NB ve... × | >

Sum of prix × | >

CA × | >

+Add data

2

Matrix

Cat	CA
Caméra	6'790'551
Jeux vidéo	4'121'711
Ordinateur	7'311'023
Mini PC	2'319'817
Ordinateur de bureau	828'246
Ordinateur portable	1'567'650
Tablette	2'155'596
Tout-en-un	439'714
Smartphone	7'362'838
Premier prix	1'137'207
Smartphone 2*	1'997'691
Smartphone 3*	1'831'062
Téléphone de jeu	1'031'868
Téléphone pliable	1'365'010
Total	25'586'123

Rows

Cat Hierarchy × | >
Cat × | >
Souscat × | >

+Add data

Columns

+Add data

Values

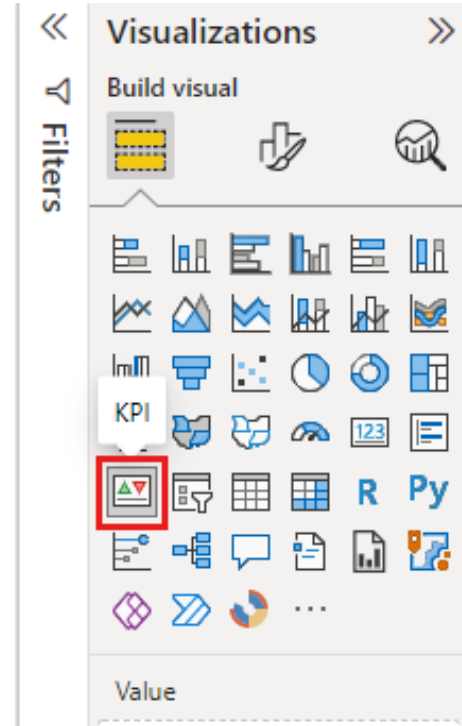
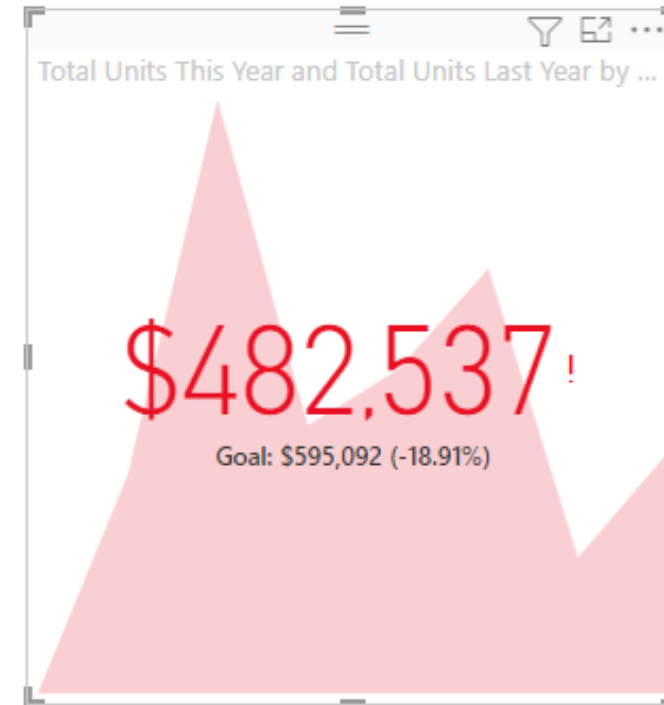
CA × | >

+Add data

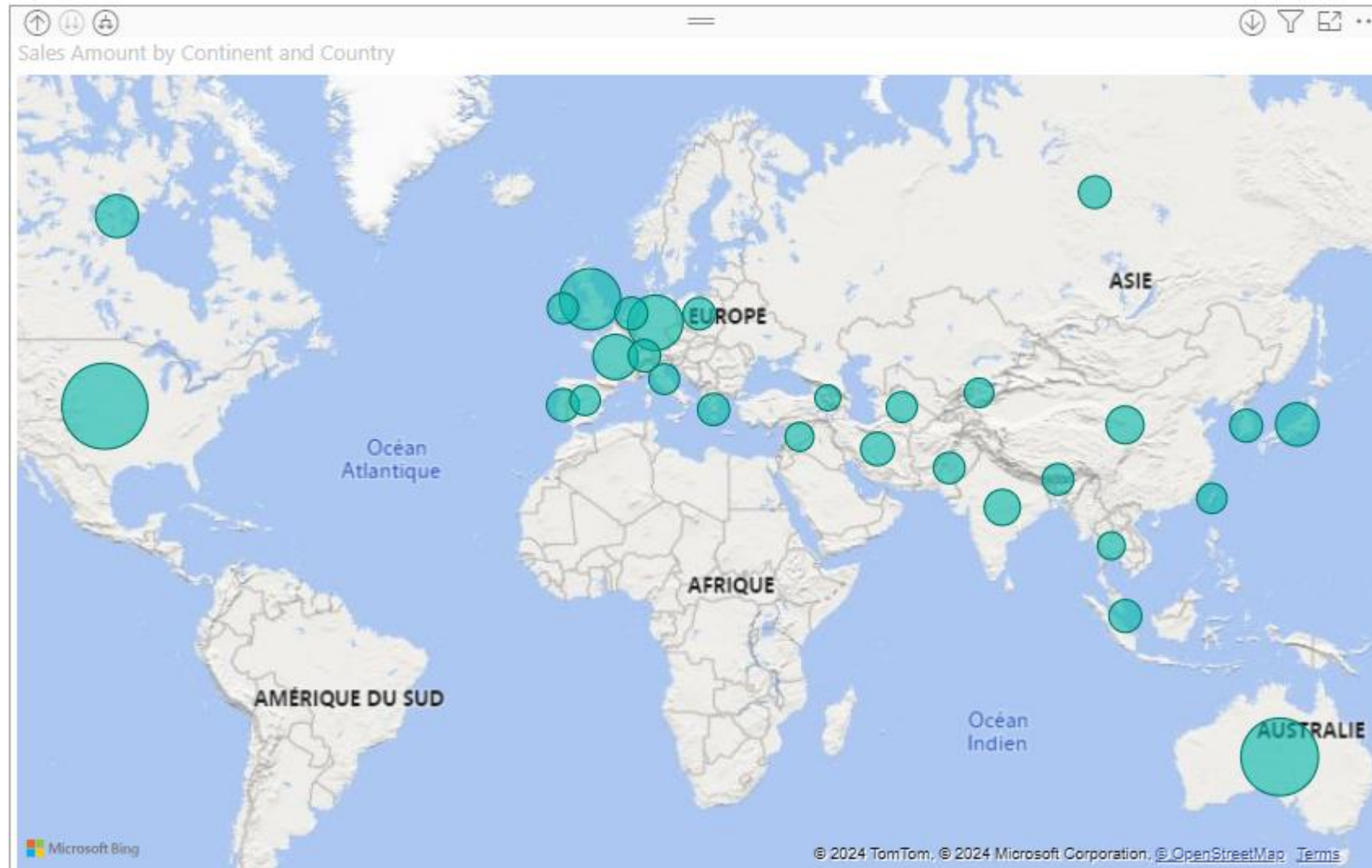


Multi-row cards

67'765'085 CHF CA	22'080'647 CHF CA 2021
22'087'219 CHF CA 2020	23'597'219 CHF CA 2022



Map visual

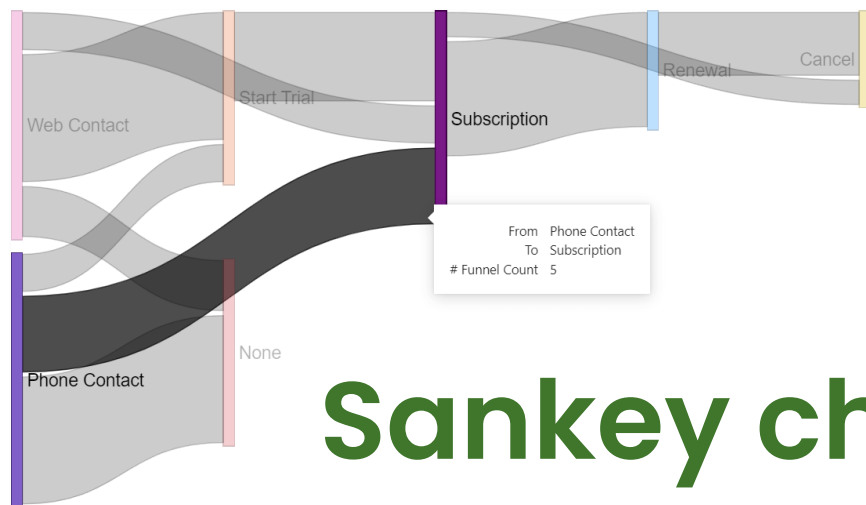
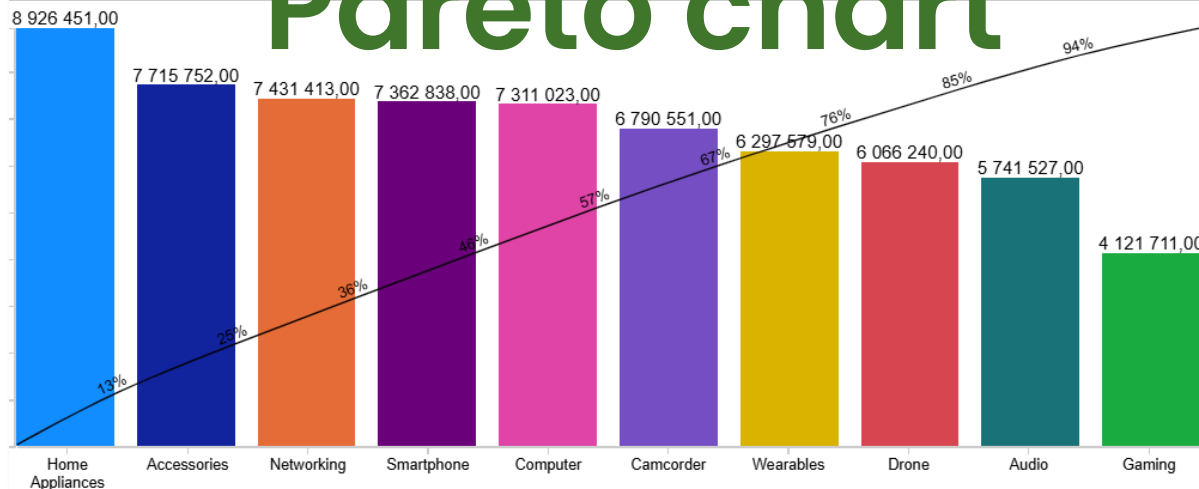


Add a new visualization



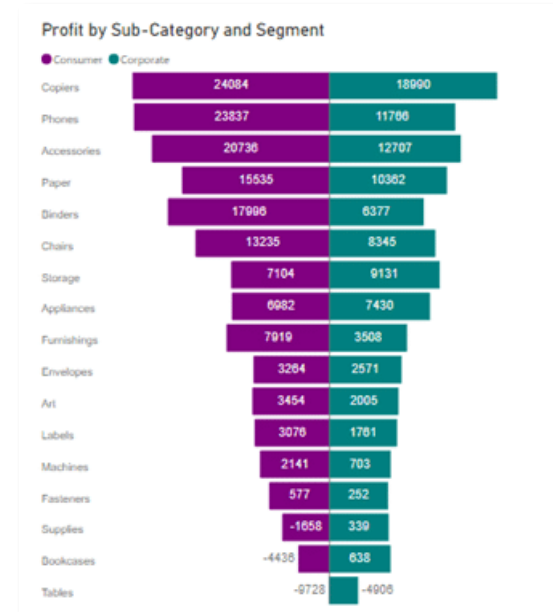
3rd party visuals

Pareto chart



Sankey chart

Tornado Chart



[More 3rd party visuals here](#)

Conditional formatting

TABLE: Top 10 weather states with afford

State	Weather	Affordability	Overall rank
Hawaii	1	45	10
Florida	2	25	5
Louisiana	3	29	36
Texas	4	24	17
Georgia	5	19	28
Mississippi	6	6	19
Alabama	7	10	16
South Carolina	8	27	41
Arkansas	9	4	11
Arizona	10	33	38

TABLE: Top 10 weather states with afford

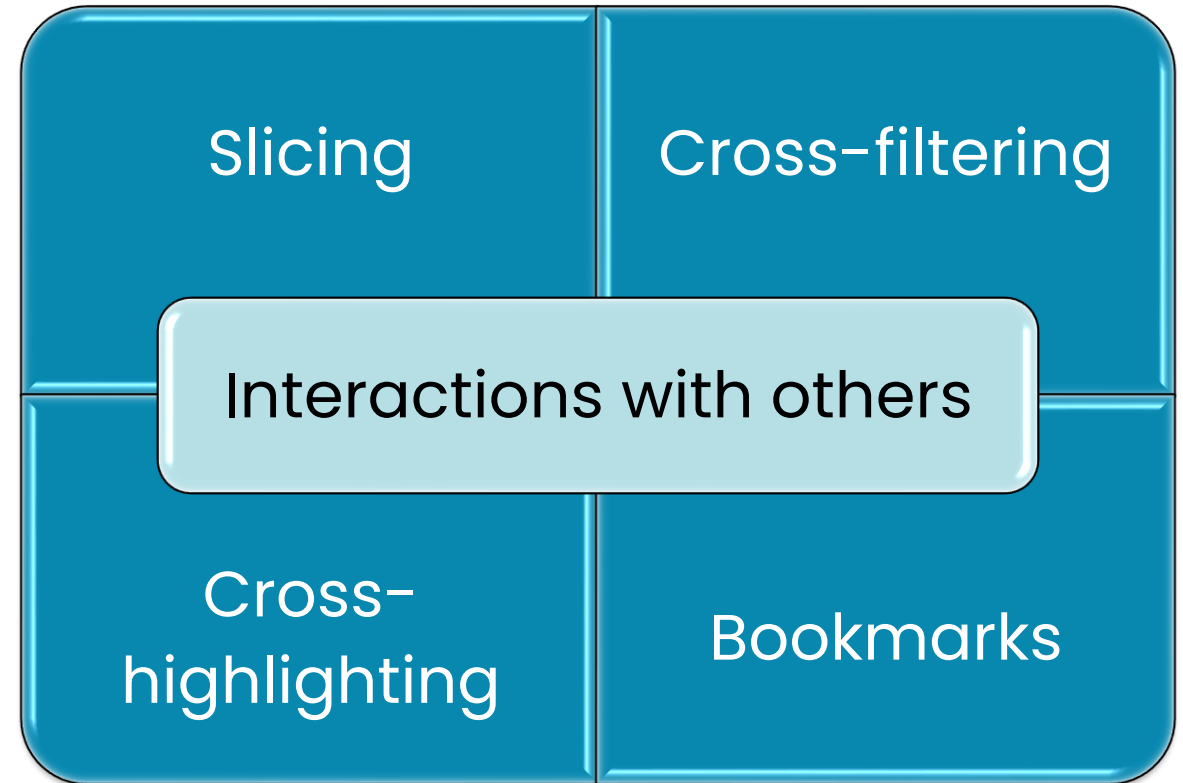
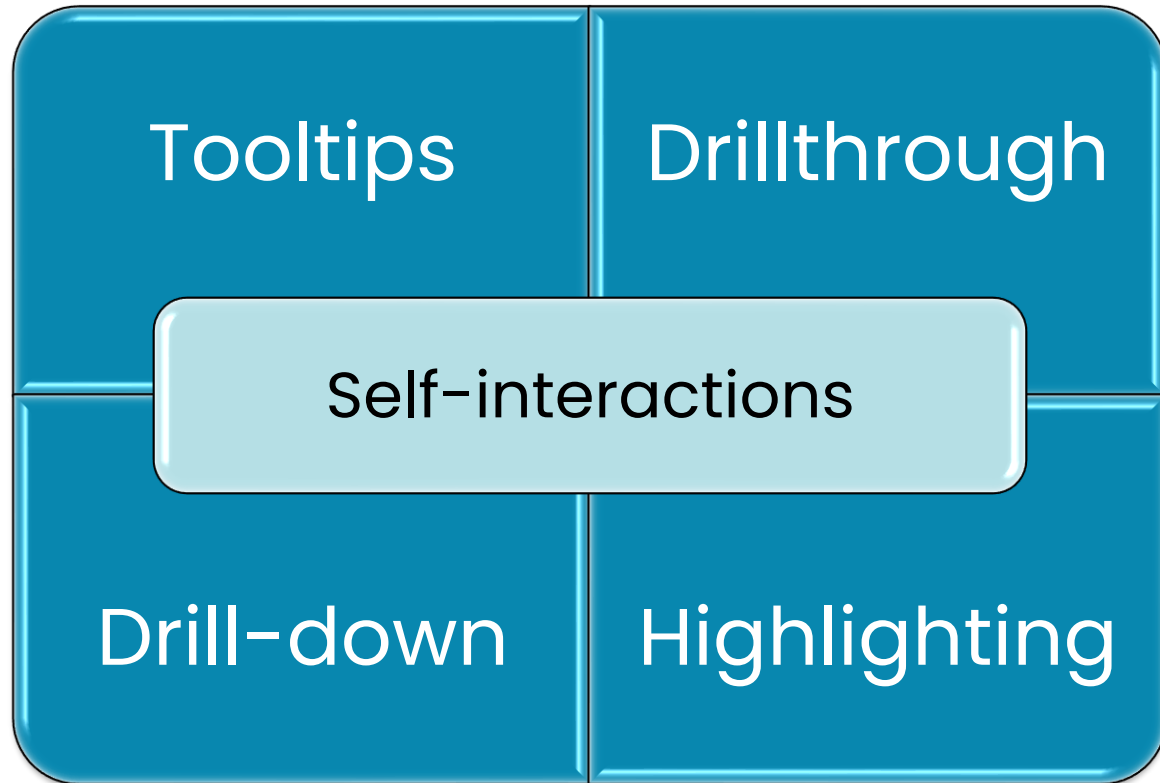
State	Affordability	Weather	Overall rank
Hawaii	45	1	10
Florida	25	2	5
Louisiana	29	3	36
Texas	24	4	17
Georgia	19	5	28
Mississippi	6	6	19
Alabama	10	7	16
South Carolina	27	8	41
Arkansas	4	9	11
Arizona	33	10	38

TABLE: Top 10 weather states with affordabil

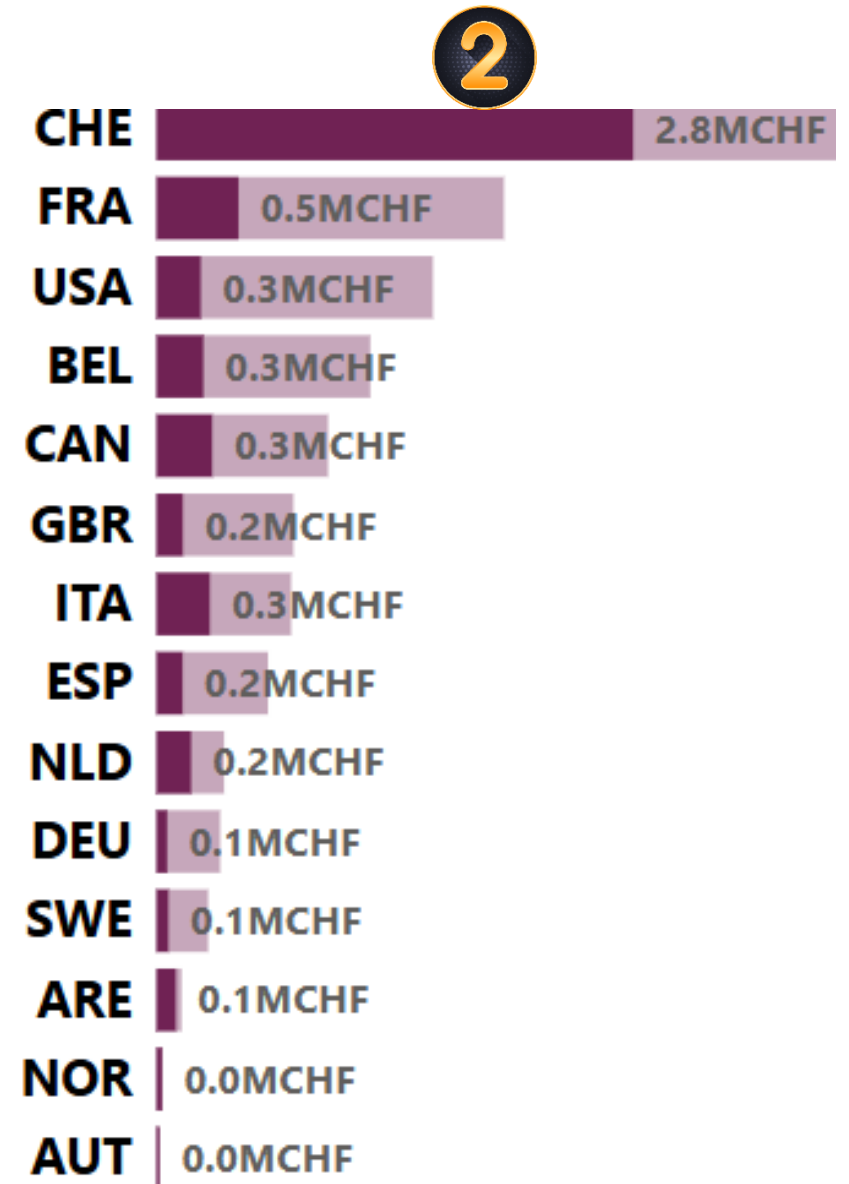
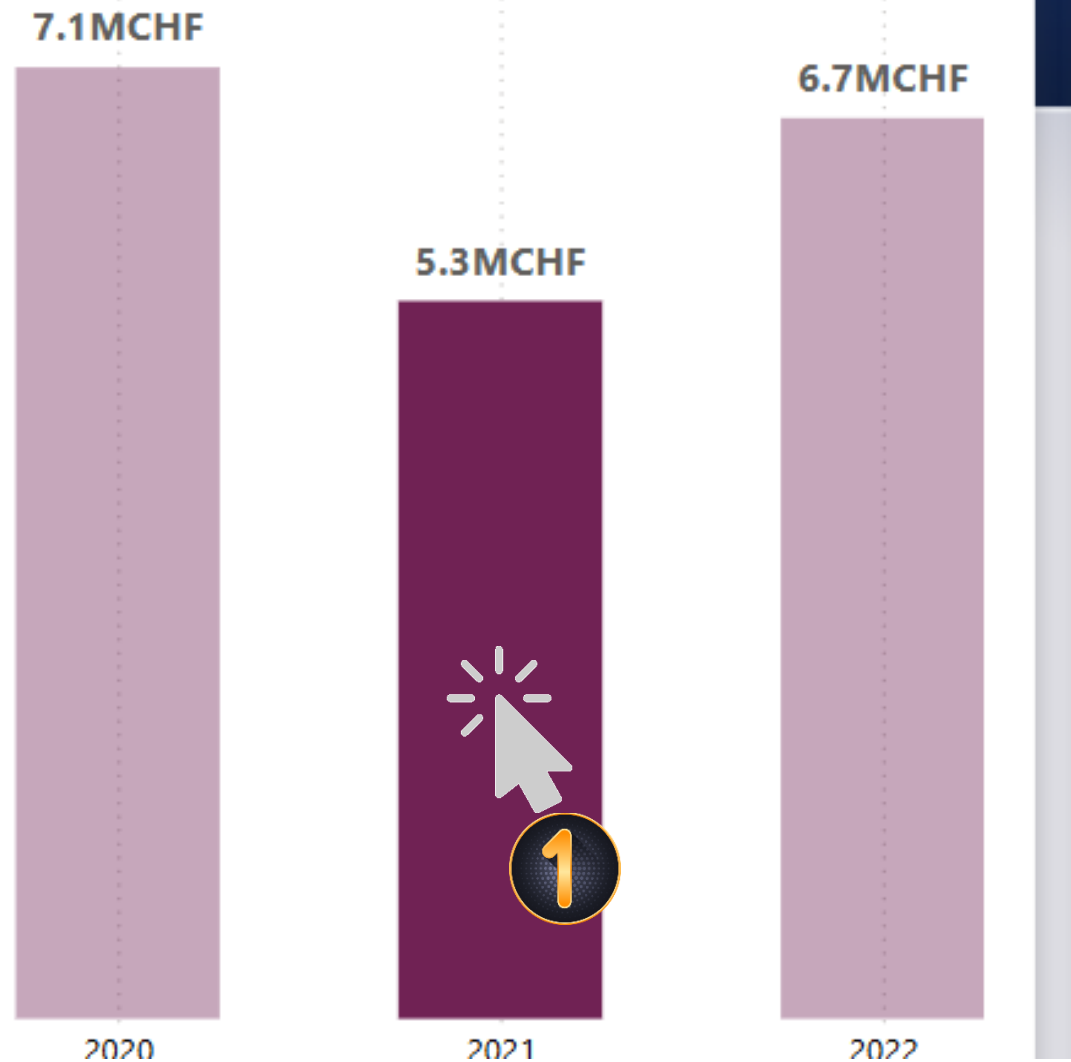
State	Affordability	Weather	Overall rank
Hawaii	45	1	10
Florida	25	2	5
Louisiana	29	3	36
Texas	24	4	17
Georgia	19	5	28
Mississippi	6	6	19
Alabama	10	7	16
South Carolina	27	8	41
Arkansas	4	9	11
Arizona	33	10	38

Interactions

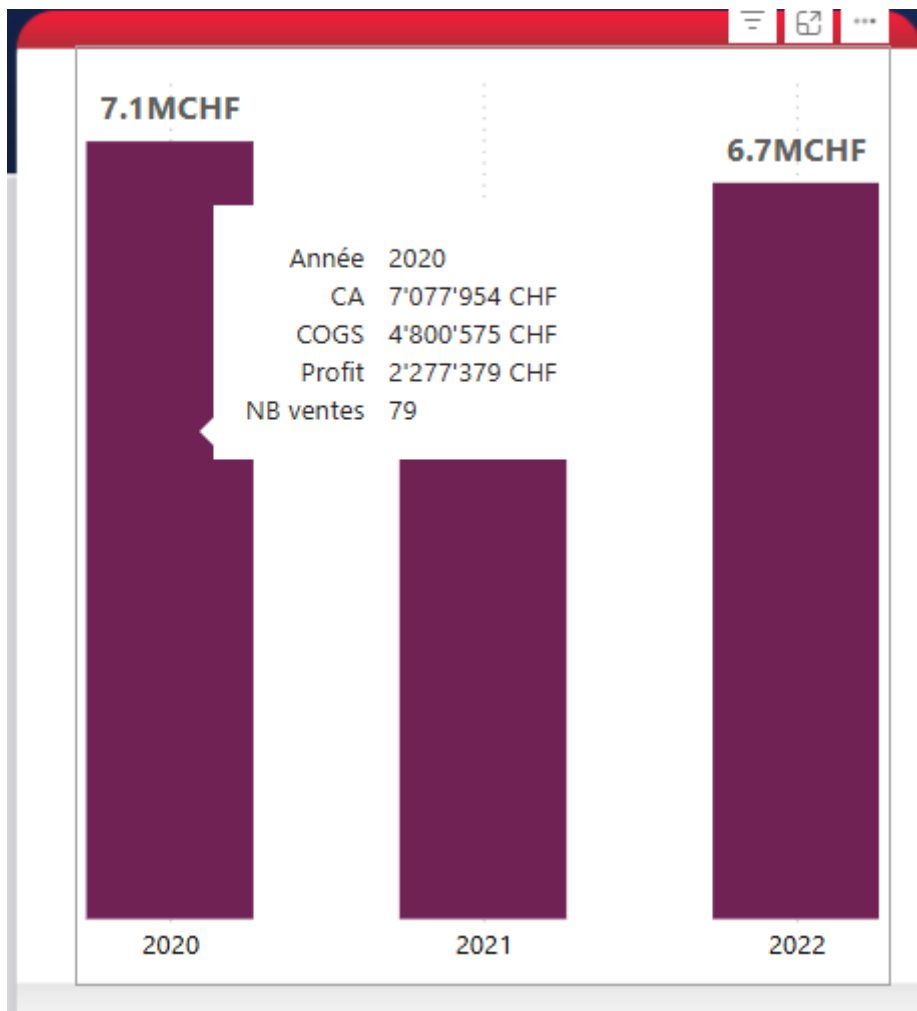
Refers to the way visuals interact with themselves or with others



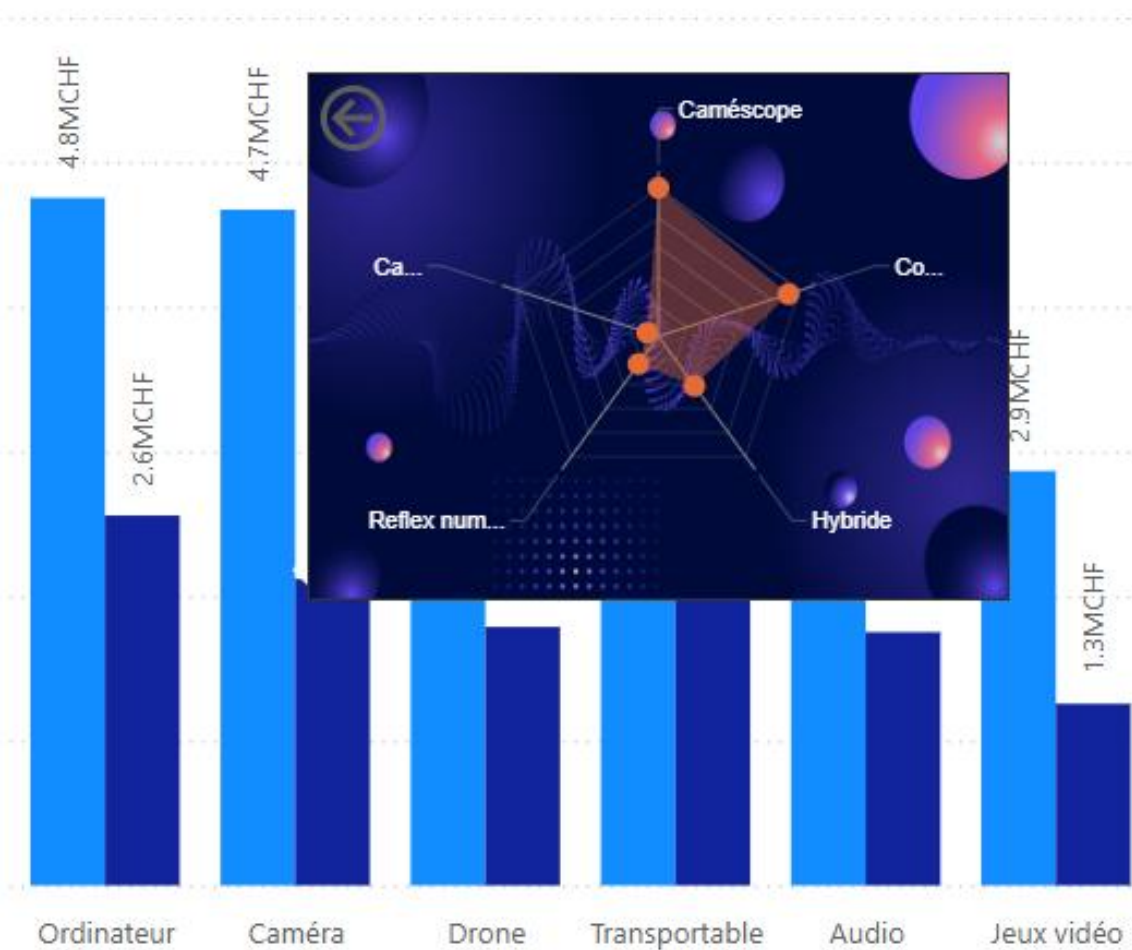
VISUAL FILTERS: Cross-highlighting



Classic tooltip

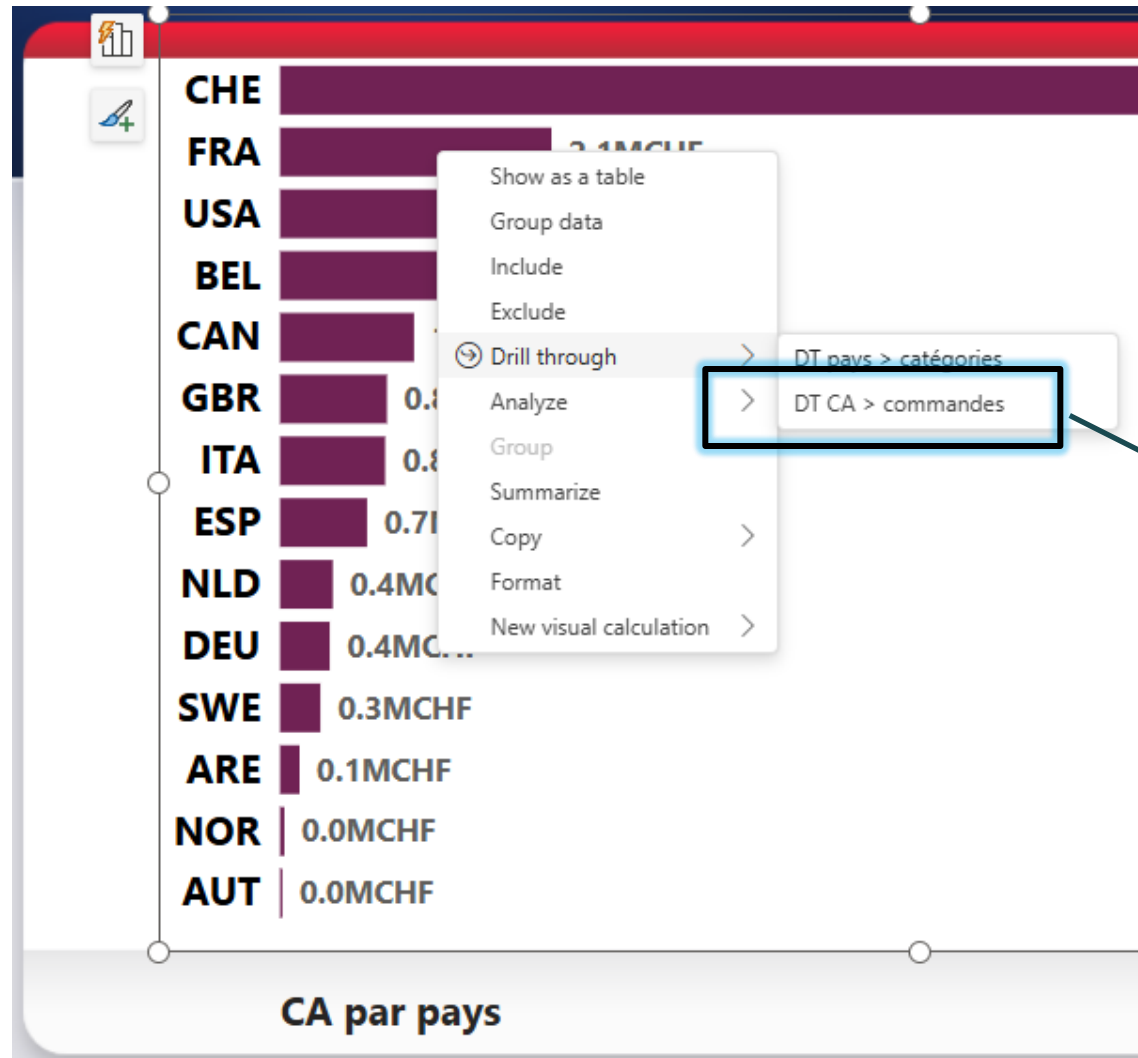


Custom tooltip



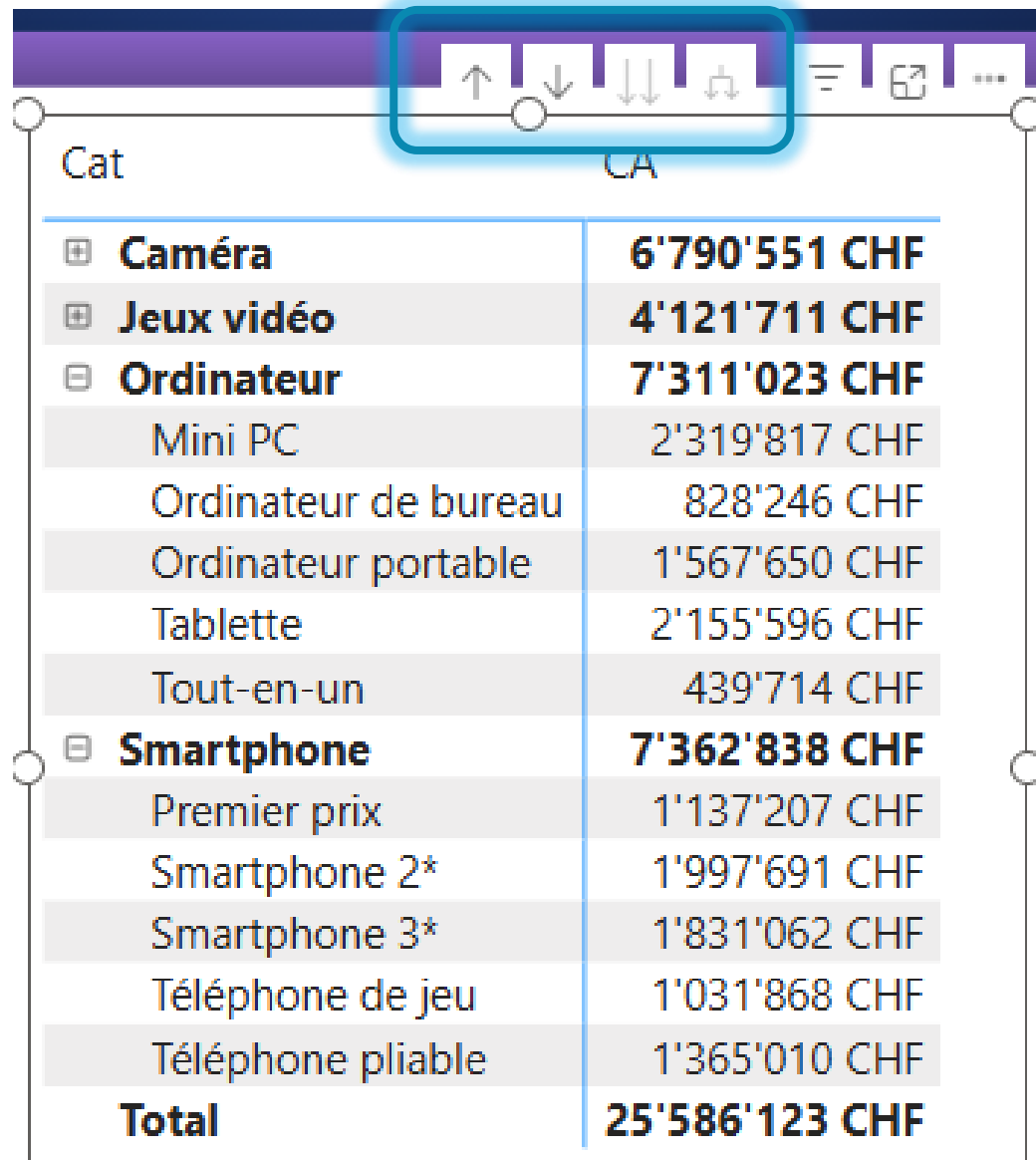
Drillthrough page

Visual in a page



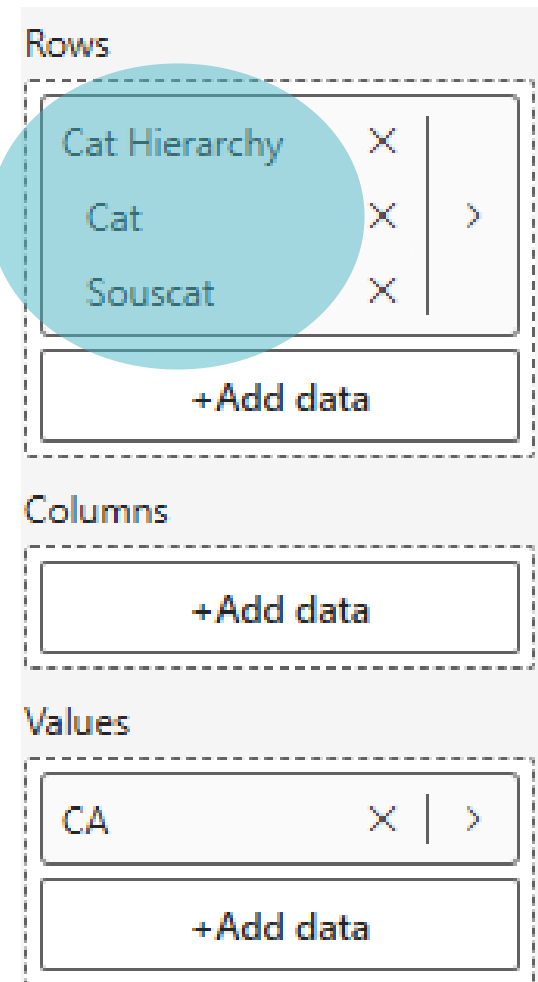
date commande	date livraison	commandeID
samedi 4 janvier 2020	samedi 11 janvier 2020	20-0001
dimanche 19 janvier 2020	samedi 25 janvier 2020	20-0010
samedi 25 avril 2020	mercredi 29 avril 2020	20-0076
dimanche 26 avril 2020	mercredi 29 avril 2020	20-0077
lundi 18 mai 2020	lundi 25 mai 2020	20-0091
lundi 8 juin 2020	samedi 13 juin 2020	20-0107
mardi 16 juin 2020	mardi 23 juin 2020	20-0113
mercredi 17 juin 2020	dimanche 21 juin 2020	20-0116
jeudi 30 juillet 2020	dimanche 2 août 2020	20-0145
lundi 12 octobre 2020	lundi 19 octobre 2020	20-0195
dimanche 17 janvier 2021	vendredi 22 janvier 2021	21-0259
mardi 16 février 2021	vendredi 19 février 2021	21-0276
lundi 15 mars 2021	lundi 22 mars 2021	21-0298
dimanche 27 juin 2021	jeudi 1 juillet 2021	21-0373
mercredi 18 août 2021	lundi 23 août 2021	21-0412
jeudi 2 septembre 2021	mardi 7 septembre 2021	21-0417
jeudi 28 octobre 2021	mercredi 3 novembre 2021	21-0456
mardi 16 novembre 2021	lundi 22 novembre 2021	21-0470
samedi 8 janvier 2022	mardi 11 janvier 2022	22-0001
dimanche 30 janvier 2022	samedi 5 février 2022	22-0019
dimanche 6 février 2022	vendredi 11 février 2022	22-0025

Drill down with hierarchies



The screenshot shows a data table with a toolbar at the top. The toolbar contains several icons: an up arrow, a down arrow, a double down arrow, a refresh icon, a filter icon, a link icon, and a more options icon. A blue box highlights the first four icons (up, down, double down, and refresh). The table has two columns: 'Cat' and 'CA'. The 'Cat' column lists various product categories, and the 'CA' column shows their corresponding values in CHF. The table is structured as follows:

Cat	CA
Caméra	6'790'551 CHF
Jeux vidéo	4'121'711 CHF
Ordinateur	7'311'023 CHF
Mini PC	2'319'817 CHF
Ordinateur de bureau	828'246 CHF
Ordinateur portable	1'567'650 CHF
Tablette	2'155'596 CHF
Tout-en-un	439'714 CHF
Smartphone	7'362'838 CHF
Premier prix	1'137'207 CHF
Smartphone 2*	1'997'691 CHF
Smartphone 3*	1'831'062 CHF
Téléphone de jeu	1'031'868 CHF
Téléphone pliable	1'365'010 CHF
Total	25'586'123 CHF



The screenshot shows a configuration panel for a data visualization. It has three sections: Rows, Columns, and Values. Each section contains a list of items with a close button (X) and a right arrow (>). The Rows section is highlighted with a blue circle. The Columns and Values sections also have a '+Add data' button below them.

Rows

- Cat Hierarchy X >
- Cat X >
- Souscat X >
- +Add data

Columns

- +Add data

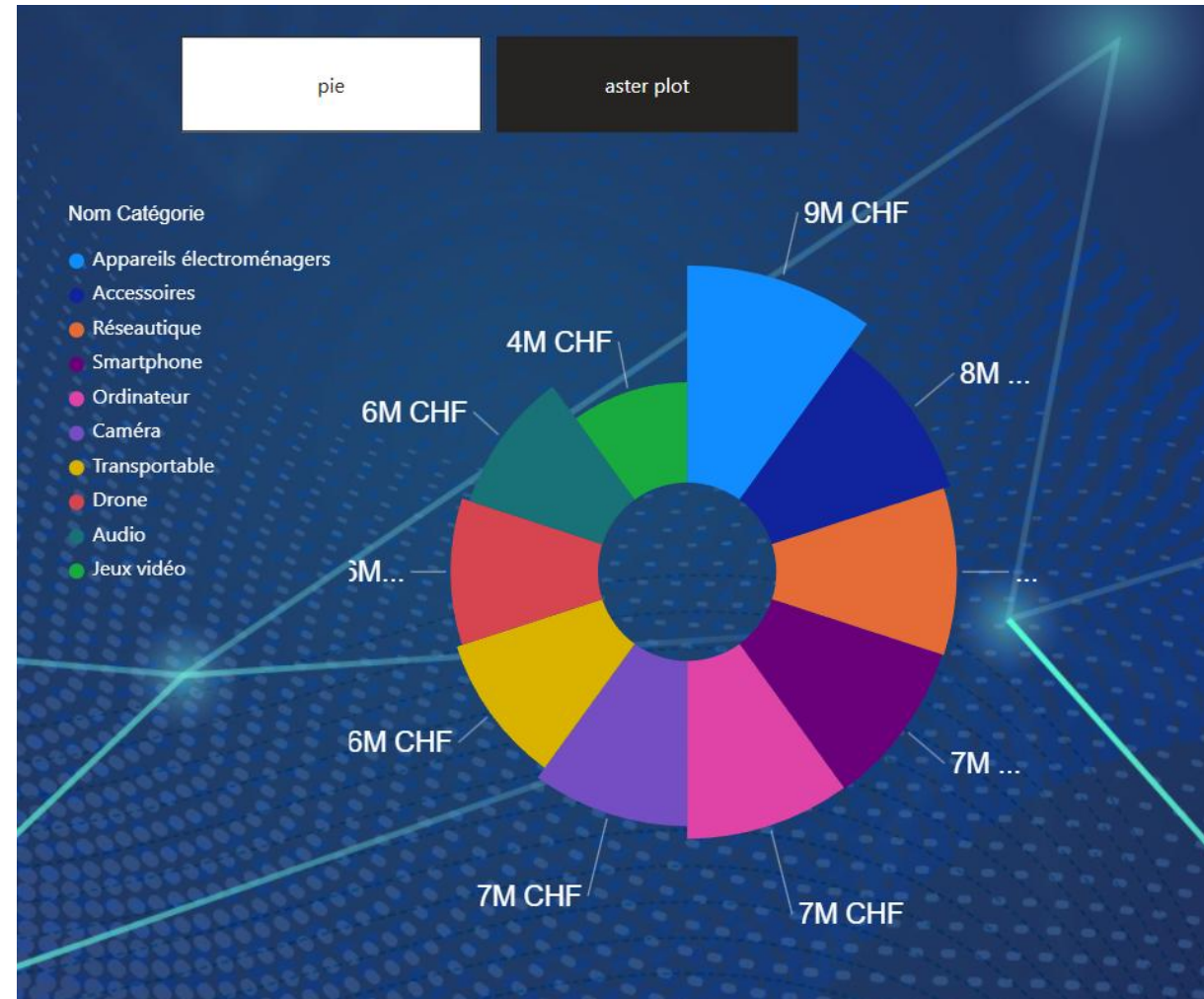
Values

- CA X >
- +Add data

Bookmarks

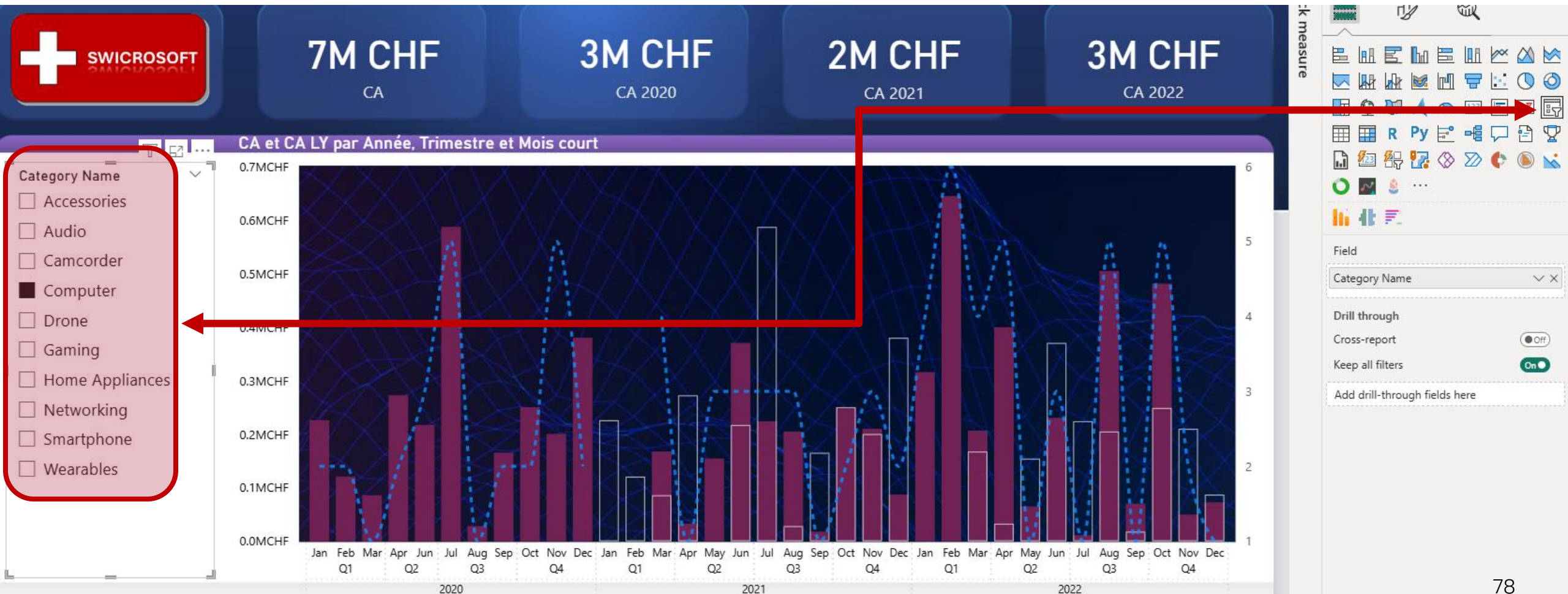
Bookmarks in Power BI are used to capture and save these specific elements of a report:

the current page, filters, slicers type and state and visual selection state. You can set up and navigate from bookmark to bookmark (as in a slideshow)



VISUAL FILTERS: SLICERS

Slicer = visual filter



Sync slicers

View ribbon / Sync slicers

The screenshot shows the Power BI ribbon with the 'View' tab selected. The 'Data / Drill' group contains the 'Sync slicers' icon. A blue arrow points from this icon to the 'Sync slicers' pane. The pane has a title bar with a close button. Below the title bar, it says 'Add and sync with all pages, or select specific pages:'. There is a table with three columns: 'Page name', a refresh icon, and an eye icon. The rows are 'Summary', 'Turnover', and 'Volume'. The 'Summary' and 'Turnover' rows have yellow checkmarks in both the refresh and eye columns. The 'Volume' row has empty checkboxes. At the bottom of the pane is a section titled 'Advanced options' with a downward arrow.

Page name	Refresh	Display
Summary	✓	✓
Turnover	✓	✓
Volume	☐	☐

Sync slicers

Add and sync with all pages, or select specific pages:

Page name

Summary

Turnover

Volume

Advanced options

Sync slicer with multiple pages

Display slicer on multiple pages

Best Practices in report view

- Within the same company, consider using a corporate PBI template (.pbit)
- Be careful with flashy background images
- Use colors with caution and with a purpose (also consider color blind people)
- Insert slicers in smart places on your report
- Write a clear and explicit title to all your visuals
- Use Drill-through and tooltip pages to add depth to your visuals
- Prefer visuals over tables and matrices (preferably certified ones)
- If you use tables , don't forget conditional formatting
- Optimize visual interactions, cross-filtering is preferred over cross-highlighting



POWER BI SERVICE

Dashboards and sharing



Power BI
Desktop



Publish to Power BI Service

The screenshot shows the Power BI Desktop interface with the 'Home' tab selected. A bar chart titled 'Sum of Sales by Customer Segment' is displayed, showing sales for Corporate, Home Office, Small Business, and Co. segments. A 'Publish to Power BI' dialog box is open, prompting the user to 'Select a destination'. The 'My workspace' option is highlighted. A 'Publishing to Power BI' success message is also visible, indicating that the report 'superstore.pbix' has been successfully published to the Power BI Service. The message includes links to 'Open 'superstore.pbix' in Power BI' and 'Get Quick Insights'. A 'Did you know?' section provides information about creating a portrait view for mobile phones. The 'Publish' button in the ribbon is circled with a blue '1'. The 'Select' button in the 'Publish to Power BI' dialog is circled with a blue '2'. The 'Got it' button in the 'Publishing to Power BI' message is circled with a blue '3'.

1

2

3

File **Home** Insert Modeling View Optimize Help

Paste Cut Copy Format painter Clipboard

Get data Excel workbook OneLake data hub SQL Server Enter data Datasource Recent sources

Transform data Refresh data New visual Text box More visuals New measure Quick measure Sensitivity Publish

Sum of Sales by Customer Segment

Count of Customers by Region

Sum of Sales

Customer Segment

Region

Central

East

Corporate

Home Office

Small Business

Co

Return Status

Normal

Returned

Publish to Power BI

Select a destination

My workspace

Publishing to Power BI

✓ Success!

[Open 'superstore.pbix' in Power BI](#)

[Get Quick Insights](#)

Did you know?

You can create a portrait view of your report, tailored for mobile phones. On the **View** tab, select **Mobile Layout**. [Learn more](#)

Got it

Select Cancel

Keep all filters

Add drill-through

Power BI service

Power BI My workspace

My workspace

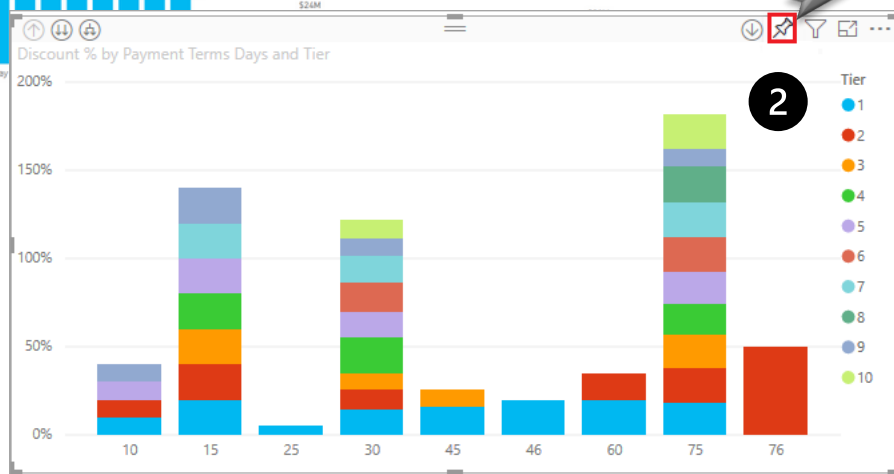
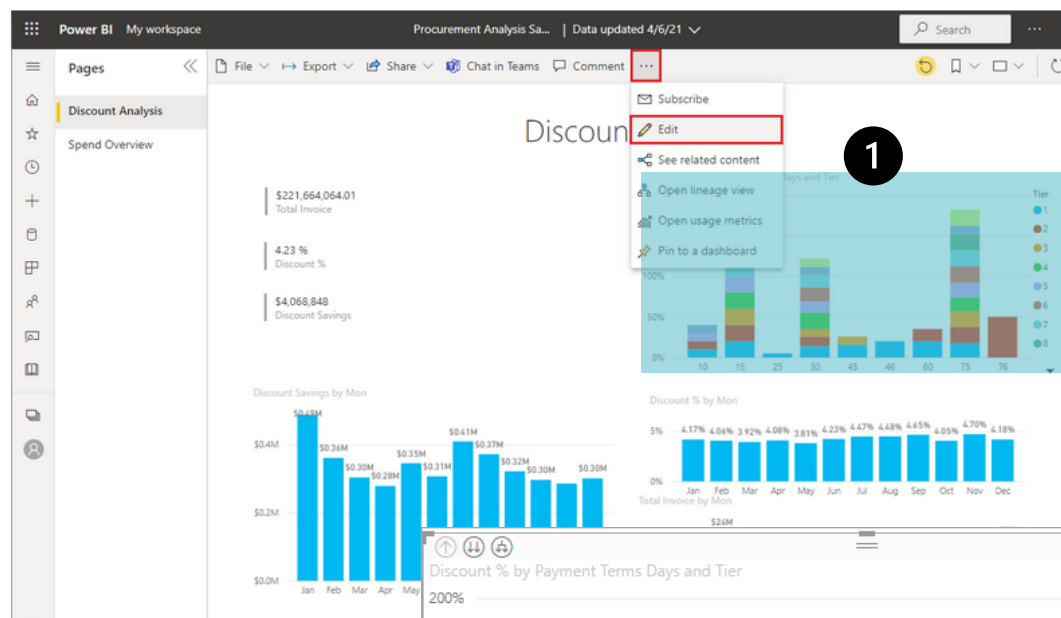
+ New item New folder → Import

	Name	Type
	Excel files	Folder
	001.MPBI01-SwiCrosoft	Report
	001.MPBI01-SwiCrosoft	Semantic model
	001.Swicrosoft	Dashboard
	AliveDX	Dashboard
	CatRegionProfit_pgrpt	Paginated Report
	Financial Analysis	Dashboard
	Financial analysis - Profit & Loss Case study	Report

PBI Service

- PBI Service hosts your published reports, semantic models and dashboards inside workspaces. You can edit reports, set alerts or subscribe to content.
- You can share content with colleagues, inside Teams or into Powerpoint.
- Dashboards are created in power BI service (not desktop). They can contain report visualizations, videos, audio, text boxes and web content.
- You can also comment (start a conversation), share or subscribe to email alerts regarding dashboards.

Adding a visual to a dashboards



Discount %
BY PAYMENT TERMS DAYS, TIER

100%
0%

10 15 25 30 45 46

Tier

1
2
3

Pin to dashboard

Select an existing dashboard or create a new one.

Where would you like to pin to?

☐ Existing dashboard

☒ New dashboard

Dashboard name

Procurement Discount %

Tile Theming

☐ Use destination theme

☒ Keep current theme

3

Pin

Cancel

Pinned to dashboard

The visualization has been pinned to your dashboard.
You can now create a phone view to optimize your dashboard for mobile phones as well.

4

Create phone view

Go to dashboard

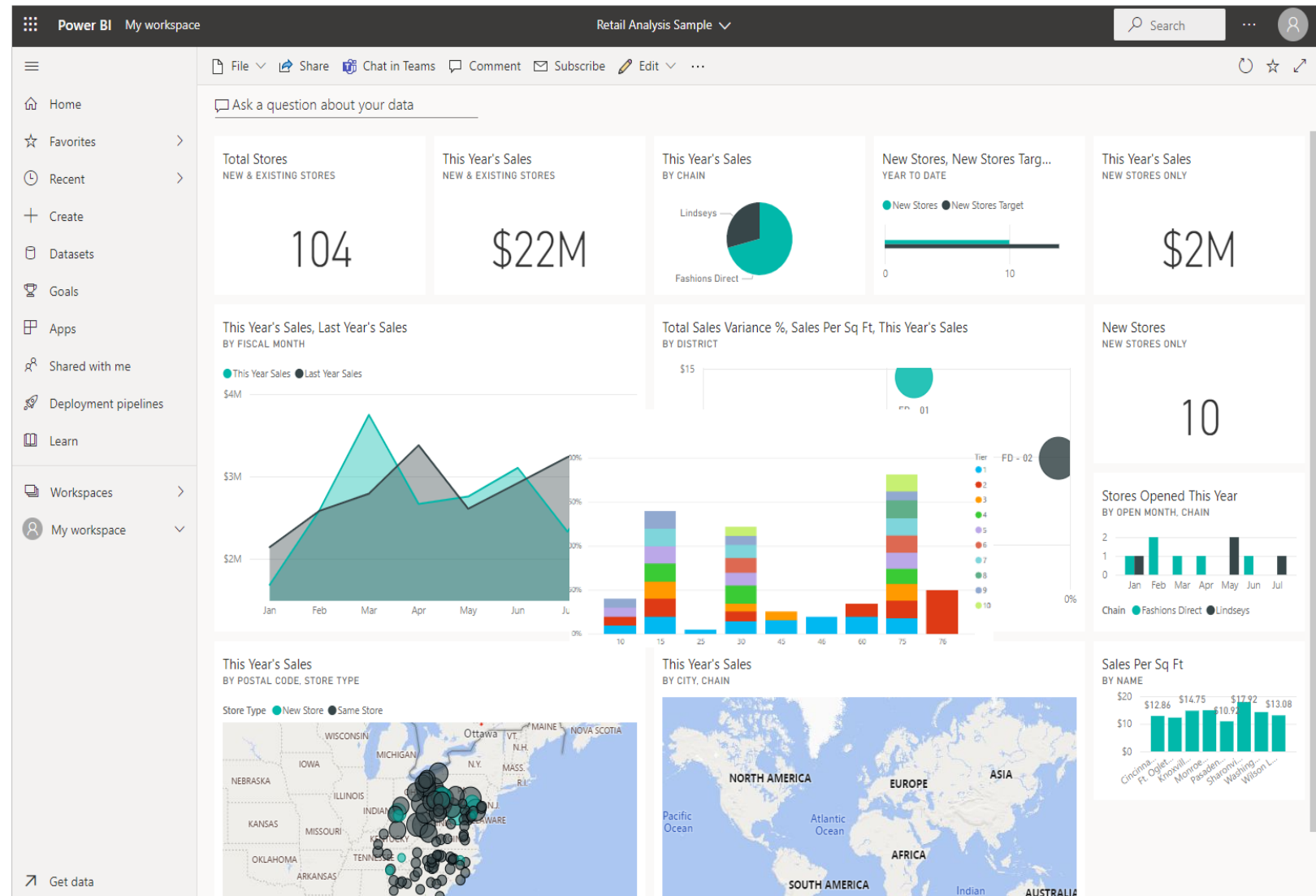
Add an entire report page to a dashboard

The screenshot shows a Power BI report page titled '001.MPBI01-SwiCrosoft' with data updated on 2/17/25. The left sidebar contains a 'Pages' pane with the 'Decomposition tree' selected. The main area displays a decomposition tree for 'Home Appliances' and 'Smart TV'. A context menu is open over the 'Smart TV' category, highlighting the 'Pin to a dashboard' option.

Category Name	Subcategory Name	Value (CHF)
Home Appliances	Smart TV	1,045,035 CHF
Home Appliances	Computer	2,557,444 CHF
Home Appliances	Networking	2,517,438 CHF
Smart TV	Home Hub	678,923 CHF
Smart TV	Smart Thermostat	579,673 CHF
Switzerland		267,710 CHF
United States of		141,330 CHF

Dashboard completed (PBI Service)

- Duplicate and rename a DB.
- Share a DB.
- Create a 'mobile friendly' version.
- Move, rename, resize, or delete tiles.
- Add new tiles from the DB menu, a Q&A or a visual inside a report page.
- Add an entire page of a report.



Sharing a report in PBI Service

My workspace

+ New item New folder → Import Migrate

	Name	Type
Folder icon	Excel files	Folder
Report icon	001.MPBI01-SwiCrosoft	Report
Semantic model icon	001.MPBI01-SwiCrosoft	Semantic model
Dashboard icon	001.Swicrosoft	Dashboard

The share icon (a square with a diagonal arrow) for the report '001.MPBI01-SwiCrosoft' is highlighted with a yellow box. A 'More options' tooltip is visible above the star icon in the same row.

Send link 001.MPBI01-SwiCrosoft

People in your organization with the link can view and share

Enter a name or email address

Add a message (optional)

Send

Copy link Mail Teams PowerPoint

Sharing a dashboard in PBI service

Power BI My workspace

My workspace

+ New item New folder Import Migrate

	Name	Type
	Excel files	Folder
	001.MPBI01-SwiCrosoft	Report
	001.MPBI01-SwiCrosoft	Semantic model
	001.Swicrosoft	Dashboard

The share icon (a square with a diagonal arrow) for the '001.Swicrosoft' dashboard is highlighted with a yellow box.

Share dashboard 001.Swicrosoft

Enter a name or email address

- ☒ Allow recipients to share this dashboard
- ☒ Allow recipients to build content with the data associated with this dashboard
- ☒ Send an email notification

Add a message (optional)

Grant access Cancel

Sharing a workspace in PBI service

Power BI PBICGL

Search

Home Create Browse OneLake catalog

POWER BI training

+ New item New folder Import Migrate

Create app Manage access Workspace settings

Filter by keyword Filter

Name	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
PBICGL	Report	—	PBICGL	8/31/2025, 10:15:08 AM	—	—	—	No
PBICGL	Semantic model	—	PBICGL	8/31/2025, 10:15:08 ...	N/A	—	—	

Manage access PBICGL

+ Add people or groups

Search within workspace

Ludovic MS Business account (i) Admin

Add people PBICGL

Admins, members, and contributors have edit and view access. Viewers only have view access. [Learn more](#)

Enter name or email

Viewer Add

Create an app in PBI service

Power BI PBICGL

Search

Home Create Browse OneLake catalog

PBICGL POWER BI training

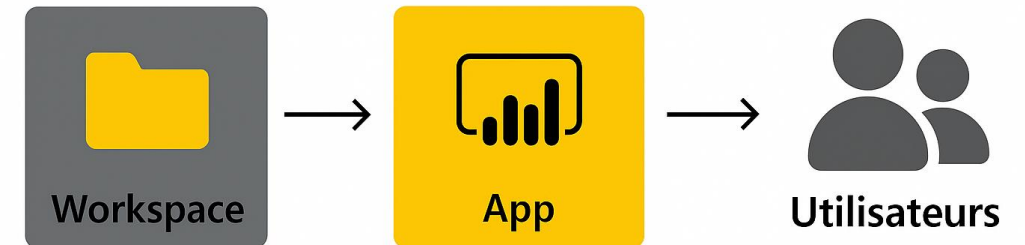
+ New item New folder Import Migrate

Create app Manage access Workspace settings

Filter by keyword Filter

Name	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
PBICGL	Report	—	PBICGL	8/31/2025, 10:15:08 AM	—	—	—	☐ No
PBICGL	Semantic model	—	PBICGL	8/31/2025, 10:15:08 ...	N/A	—	—	

Une App est un moyen de regrouper et partager des rapports, dashboards et datasets dans un package unique, facile à distribuer aux utilisateurs ou groupes dans votre organisation.



Pourquoi créer une App ?

- Centraliser plusieurs rapports et tableaux de bord dans un seul espace.
- Contrôler les accès (sécurité et permissions).
- Offrir une interface simple pour les utilisateurs finaux.

Power BI service and Gateways

What is a gateway in Power BI?

A gateway is a component that allows Power BI Service (in the cloud) to access on-premises data (for example, a SQL server in your company). It acts as a secure bridge between your internal network and the Power BI service.

How do I set up the Gateway?

1. Download and install the Gateway from the Power BI site.
2. Register the Gateway in Power BI Service with your account.
3. Configure data sources in Power BI Service:
 - Set the type (SQL Server, etc.).
 - Provide authentication information.
4. Associate the Gateway with datasets for scheduled updates or DirectQuery.

Gateways: Best practices

- Install the Gateway on a dedicated server
 - Avoid user workstations to ensure availability and performance.
 - Choose a server that is close to the data sources to reduce latency.
- Recommended mode: Enterprise Gateway
 - For shared reports and multiple sources.
 - Enables centralized management and high availability.
- Configure security and access
 - Use service accounts with the necessary rights to the SQL source.
 - Enable authentication through Azure Active Directory if possible.
- Plan for high availability
 - Install multiple instances in a cluster to avoid disruptions.
- Monitor and maintain the Gateway
 - Update regularly.
 - Use the Power BI Gateway Performance Monitor to track resources.



POWER BI: BEST PRACTICES



Power BI

Desktop



1.1 Data Modeling

- Use the star schema with fact tables (transactions, events) in the center and dimension tables (attributes) around it. This makes reporting more efficient and easier to understand.
- Avoid complex relationships: many-to-many relationships and circular dependencies. Use one-way relationships whenever possible.
- Use calculated columns sparingly: Use metrics for calculations instead, as calculated columns can slow down performance and consume more memory.
- Use the right data types: Assign the correct data types (dates, numbers, text) to columns to ensure accurate calculations and aggregations.

1.2 Data Cleansing and Transformation

- ◇ Use Power Query to cleanse, filter, and transform data before uploading it to Power BI. This reduces the processing overhead when interacting with the report.
- ◇ Remove unnecessary columns: Load only the columns you need for analysis. This reduces the size of the dataset and improves performance.
- ◇ Rename columns and tables for better readability and consistency.

2.1 Report Design

- Set a clear goal: Each report should have a specific purpose. Align the report design with business objectives or KPIs to ensure its relevance.
- Use a narrative approach: Guide users through the report by structuring it logically. Start with an overview (dashboard) and then drill down into the details.

2.2 Choosing visualizations

- ◇ Select appropriate visuals: Choose visuals that are appropriate for the type of data and purpose (e.g., histograms for comparisons, line charts for trends, pie charts to use sparingly).
- ◇ Limit the number of visuals per page: Too many visuals can create confusion and degrade performance. Limit yourself to 4–6 visuals per page.
- ◇ Ensure visual consistency: Use the same colors, fonts, and chart types as much as possible. Maintain consistent scales for focuses on similar visuals.
- ◇ Provide context: Add labels, titles, tooltips, and captions to help users quickly understand the data.
- ◇ Use slicers and filters carefully: Allow users to slice and filter data, but avoid overloading it with too many options.

2.3 Page layout

- ◇ Use a clean layout with enough white space between visuals to avoid clutter.
- ◇ Arrange items logically, usually from top to bottom or left to right, according to natural reading habits.
- ◇ Use themed colors that align with the organization's brand guidelines. Power BI allows you to set custom themes to ensure color consistency.

3.1 Navigation and interactivity

- ◇ Create detail pages to allow users to drill down into more detailed data without overloading the main dashboard.
- ◇ Use bookmarks and buttons to make reports interactive (for example, switching between different views or changing visuals dynamically).
- ◇ Use tooltips to provide additional information when hovering over visuals, without cluttering up the main page.

3.2 Responsive design

- ◇ Make sure the reports are responsive and adapt to different screen sizes, especially for mobile users. Power BI offers a mobile-specific layout.
- ◇ Test reports on different devices to ensure that all visuals display correctly.

4.1 Data Model Optimization

- ◇ Large dataset aggregations: When possible, aggregate data before uploading it to Power BI. For example, summarize them at the weekly or monthly level rather than the daily level.
- ◇ Use Power BI aggregations: Define aggregated tables for quick access to summary data.
- ◇ Optimize DAX calculations: Avoid complex DAX formulas that could hurt performance. Use variables to break down calculations and improve readability.

4.2 Performance improvements

- ◇ Use the Power BI performance monitor to detect and optimize slow-running visuals and queries.
- ◇ Limit visuals to a single page: Too many visuals can slow down report performance. Use calculated metrics, summaries, or smaller data sets to improve load times.
- ◇ Enable query reduction options to minimize the number of queries generated by filters and segments, which reduces the load on data sources.

5. Data Security and Sharing

- Row-level security (RLS): RLS restricts access to specific data in the report based on user roles. This ensures that users only see the data they are entitled to.
- Publish to the appropriate workspaces: Ensure that reports are published to the right Power BI workspaces based on business needs (e.g., Development, Production).
- Track report usage: Use Power BI usage metrics to understand how users interact with the report and make improvements based on their behavior.

6. Release management and documentation

- Versioning: Use versioning tools such as OneDrive or SharePoint for report files. Power BI doesn't have native version control, so tracking versions manually helps with collaboration.
- Document the data model: Provide documentation or tooltips explaining the structure of the report and key metrics, to help users understand the logic behind the visuals.

7. Testing and iterations

- TesEngage with representative users: Before finalizing the report, test it with a group of end-users to ensure that it meets their needs and performs.
- Iterate based on feedback: Power BI reports are often living documents. Gather user feedback and make iterative improvements to better meet evolving business needs.

8. General Reporting Guidelines

- Use KPIs and dashboards: Include KPIs to highlight critical metrics at a glance. Maps and dashboards are great for displaying overall performance.
- Explain variances: When showing trends or comparisons, use visual aids such as reference lines, goal indicators, or tooltips to explain significant variances.
- Avoid excessive detail: Provide only high-level information in the main dashboard and allow users to dig deeper into the data if needed.

9. Summary of Best Practices:

- Model efficient data elimination with star schema and appropriate relationships.
- Clear and meaningful visualizations aligned with the purpose of the report.
- Consistent design focused on readability and user experience.
- Optimized performance through DAX optimization, aggregation, and visual throttling.
- Secure and share with row-level security, certified datasets, and careful workspace management.



POWER BI: PRICING & other products

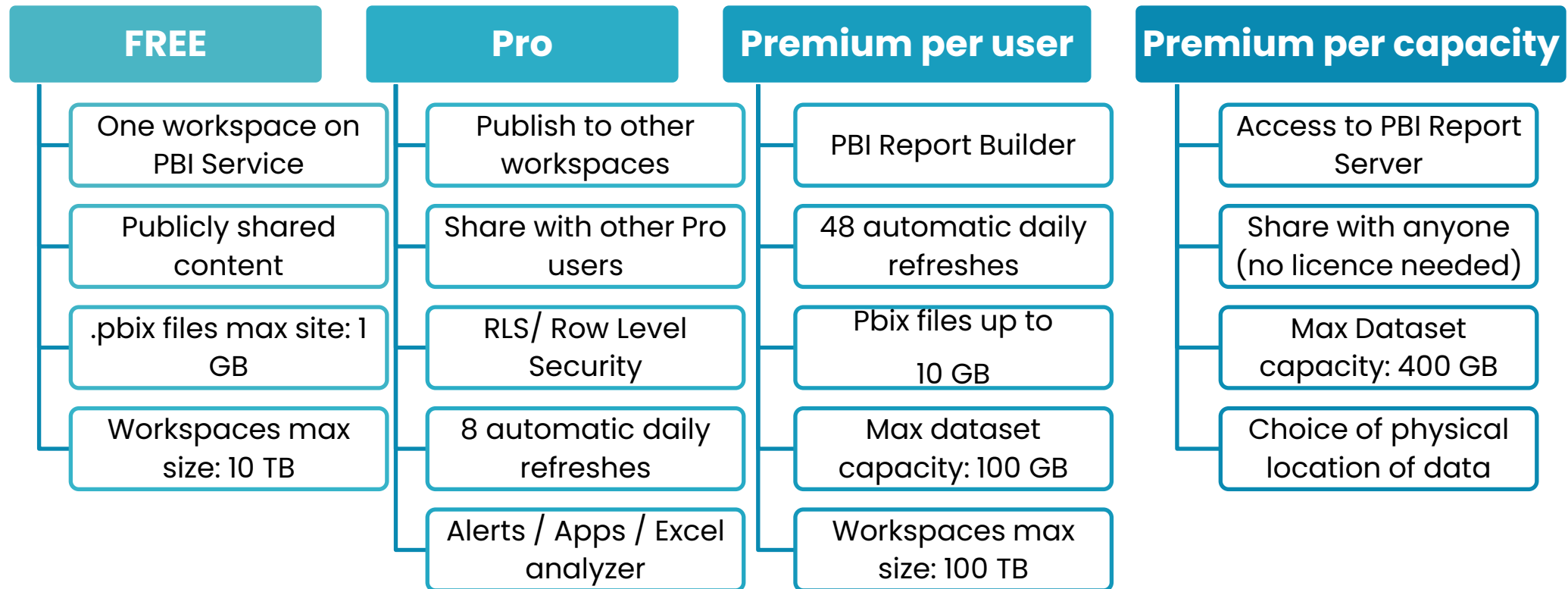


Power BI

Desktop



Power BI Licensing (<https://powerbi.microsoft.com/fr-ch/pricing/>)



The POWER BI family

Power BI Desktop

Power BI Service

Power BI Mobile

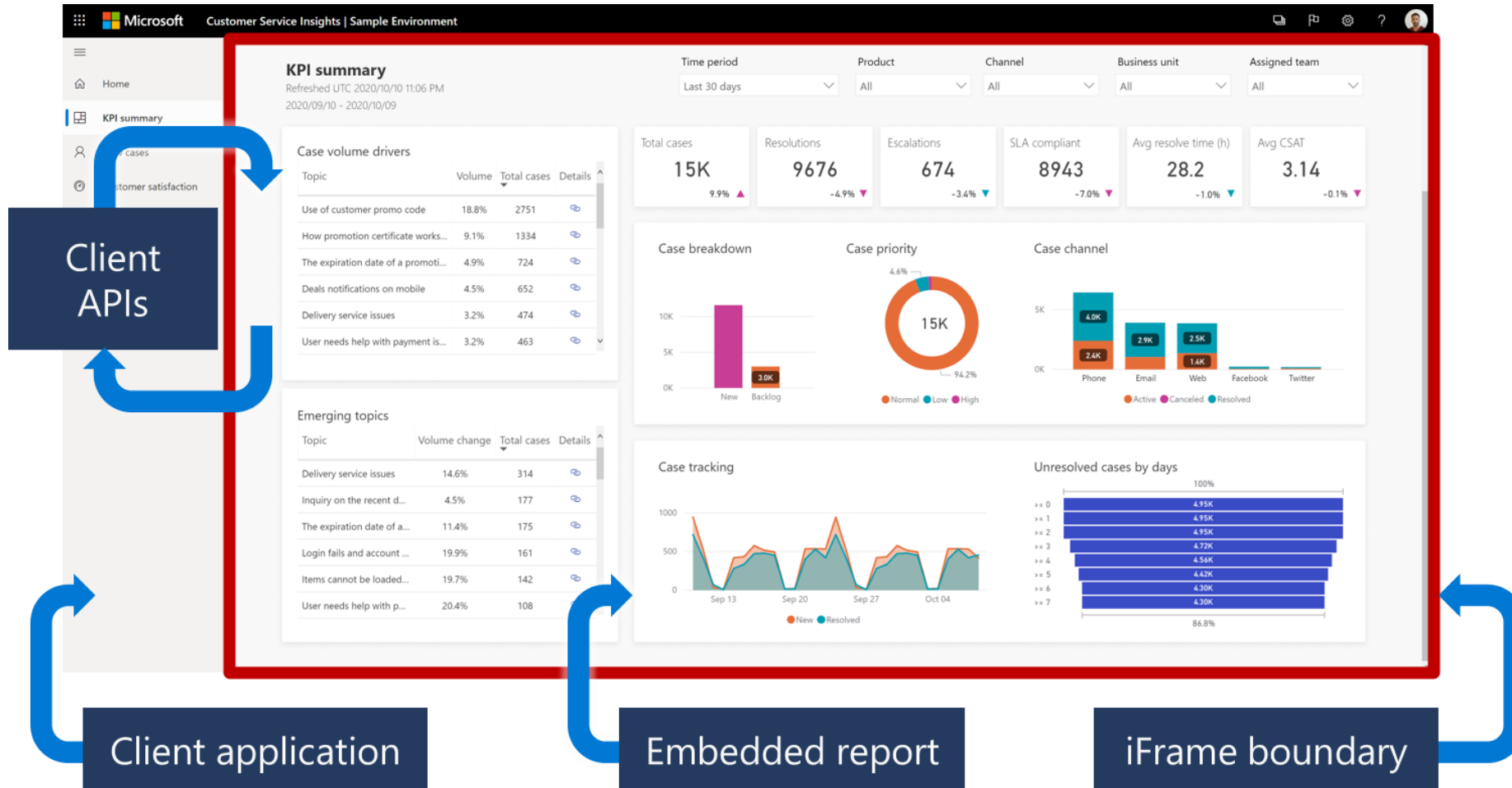
Power BI Report builder

Power BI Report Server on premises

Power BI Embedded

Power BI Embedded

is an Azure service that allows companies to integrate Power BI's analytical capabilities into other applications



Power BI Report builder

- allows you to construct and share structured, paginated reports from data.
- Generates .rdl files.

Invoice.rdl - Power BI Report Builder

File Run

Design Zoom First Previous 1 of 1 Next Last Refresh Stop Back

Views Zoom Navigation Print Page Setup Print Layout Export Parameters Document Map Options Find

Select company: Contoso Suites View Report

Contoso

Contoso Suites
123 Coffee Street
Buffalo, NY 98052
USA
Telephone 012345678
<http://www.contoso.com>

Owl Wholesales
123 Violet Road, Phoenix, CO 85003 USA

Invoice CIV-000676 000007-1
30 November 2019
Payment terms: Net 45 days
Payment due 1/14/2020
\$321,113.52

ITEM	DESCRIPTION	QUANTITY	SALES PRICE	DISCOUNT	AMOUNT
D011	Lens	2 Each	2	0	4.00
L0001	Mid-Range Speaker	35 Each	500	0	17,500.00
P0001	Acoustic Foam Panel	117 Each	37	0	4,329.00
D0003	Standard Speaker	23 Each	220	0	5,060.00
T0001	Speaker cable 10	65 Each	500	0	32,500.00
D0004	High End Speaker	12 Each	2000	0	24,000.00
T0004	Television M120 37" Silver	53 Each	350	0	18,550.00
T00002	Projector Television	23 Each	3750	0	86,250.00
T0005	Television HDTV X590 52" White	33 Each	2890	0	95,370.00
T0003	Surround Sound Receiver	56 Each	450	0	25,200.00
SALES SUBTOTAL AMOUNT					4.00
SALES TAX					12,350.52

Sales invoice notes

METHODS OF PAYMENT

Electronic payment
Payment **US-009**
Sort code
Account No. 34567

Check
Make check payable to **Contoso Suites**

OTHER INFORMATION

Tax registration no. 1234123400
Our reference **Karl Bystrom**

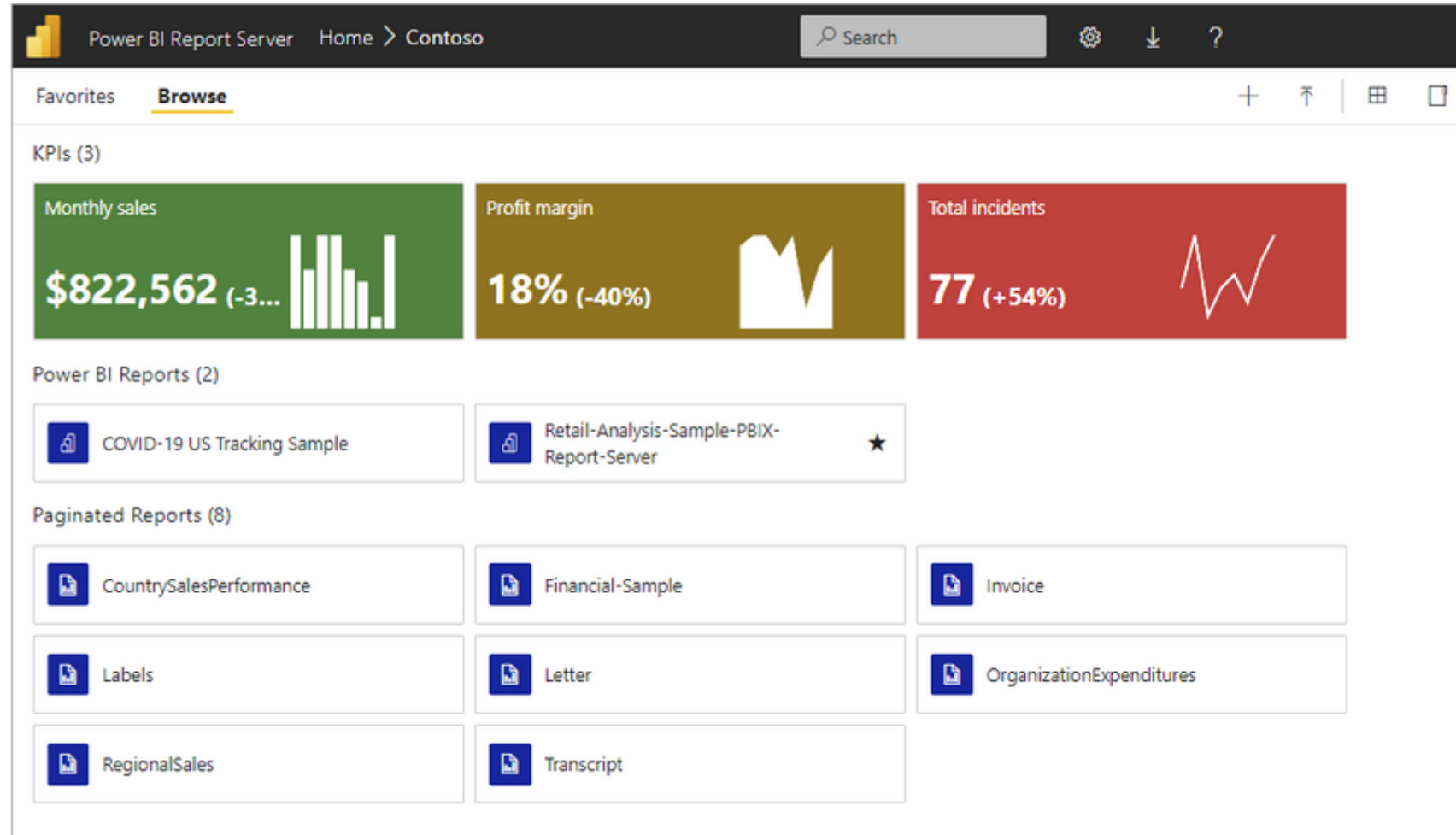
Disclaimer: This is a sample and is provided 'as is' without warranty of any kind. Sample data included are for illustration only and are fictitious. No real association is intended or inferred.

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Power BI Report Server on premises

is an on-premises report server with a web portal in which you display and manage reports and KPIs. Along with it come the tools to create Power BI reports, paginated reports, mobile reports, and KPIs.



Useful links

- Getting started with Power BI:
<https://learn.microsoft.com/en-us/power-bi/fundamentals/>
- DAX functions reference:
<https://learn.microsoft.com/en-us/dax/dax-function-reference>
- Data modeling:
<https://powerbi.microsoft.com/en-us/what-is-data-modeling/>
- Power query:
<https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-query-overview>

